Accumulated-Variance Functionals of Hidden Markov and Semi-Markov Processes: Finite-Sample Upper and Lower Bounds

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Abstract

We study two accumulated variance functionals for stationary two-state hidden Markov or hidden semi-Markov process, observed through a single trajectory, and separate identifying these observable quantities from recovering the latent parameters behind them. For bounded durations, inverting block moments gives finite-sample upper bounds, worked out with fully explicit constants for a capped-geometric Gaussian family; on fixed interior classes, the corresponding two-point lower bounds have the same rate. On prescribed local persistence cells a long-lag estimator matches the regular and saturated regimes—though only cell by cell, with a lag built around the cell's own scale, not through any single estimator that adapts across unknown scales. Unbounded durations known only through two moments are a separate matter: no uniform exponential bound is claimed there either.

Keywords: accumulated variance; hidden Markov process; hidden semi-Markov process; finite-sample concentration; minimax lower bound; persistence.

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1 Introduction

This paper concerns inference for two observable probability functionals from one stationary trajectory: the variance accumulated over a finite prediction horizon and its long-run per-unit-time limit. The data length and prediction horizon are kept separate throughout. This distinction is essential because estimation error is determined by the former, whereas the sensitivity of the target may grow with the latter.

The first structural point is that identification of the target functional is strictly weaker than identification of a latent parametrization. We therefore derive deterministic stability inequalities directly in observable moment coordinates before propagating error through a latent inverse. The resulting bounds remain meaningful at some parameter configurations where full latent recovery is unavailable, provided that the accumulated-variance functional is still identified.

The two-state Markov model serves as a benchmark: its geometric covariance structure exposes the finite-horizon sensitivity and permits a transparent three-moment estimator. The central semi-Markov result treats bounded durations through residual-time Markovization and block-moment inversion. In the capped-geometric Gaussian family, the observable inverse, a uniform identification margin, and a pseudo-spectral-gap lower bound are all calculated explicitly rather than left as abstract regularity constants.

The comparison splits in two. On fixed interior classes, a Bernstein bound for observable moments and a two-point lower bound have the same powers of data length, confidence level, and prediction horizon. Along the symmetric persistent path, a lag tied to the persistence time attains the corresponding lower-bound powers on either side of saturation—but only cell by cell, with the cell scale fixed in advance; we do not construct an estimator adaptive to an unknown persistence scale.

The computational procedures are included to show that the theoretical quantities are evaluable and to audit constants, indices, and saturation thresholds. They are not the basis of the contribution. The paper also records why two duration moments alone cannot support a uniform exponential claim for unbounded sojourn laws.

2 Observation experiment and estimands

Let $S_t \in \{1, 2\}$ be stationary and let the observations be conditionally independent given the latent path, with state-dependent emission laws Q_i and

$$\mathbb{E}(X_t \mid S_t = i) = \mu_i, \qquad \mathrm{Var}(X_t \mid S_t = i) = \sigma_i^2.$$

Only $X_{1:T} = (X_1, \ldots, X_T)$ is observed. To avoid a rate ambiguity, T always denotes the data length and N the horizon of the target

$$G_N(\theta) := \mathrm{Var}_\theta\left(\sum_{t=1}^{N} X_t\right), \qquad v_\infty(\theta) := \lim_{N\to\infty} N^{-1} G_N(\theta),$$

whenever the latter limit exists. The exact expression for $G_N(\theta)$ is taken from Lemma 1 of the population analysis and is not rederived here.

For the two-state alternating semi-Markov model, complete sojourns $D_i \sim F_i$ have probability mass f_i, mean τ_i, and variance ν_i^2. Consecutive sojourns are mutually independent. The stationary occupation probabilities and inclusive residual-life laws are

$$\pi_i = \frac{\tau_i}{\tau_1 + \tau_2}, \qquad \mathbb{P}(R_i = r \mid S_1 = i) = \frac{\mathbb{P}(D_i \geq r)}{\tau_i}, \quad r \geq 1. \tag{1}$$

The relevant aperiodicity condition is the span-one condition on a complete return cycle,

$$\gcd\{d_1 + d_2 : f_1(d_1) f_2(d_2) > 0\} = 1. \tag{2}$$

Separate span-one assumptions on F_1 and F_2 are not equivalent to (2) and are not, by themselves, sufficient.

3 Identification

Definition 1 (Latent and functional identification)**.** Let $P_\theta^{(\ell)} = \mathcal{L}_\theta(X_{1:\ell})$. A latent parameter is identified from ℓ observations, up to state permutation, when

$$P_\theta^{(\ell)} = P_{\theta'}^{(\ell)} \quad \Longrightarrow \quad \theta' = s\theta$$

for a permutation s of the two state labels. A scalar functional g is identified when the weaker implication $P_\theta^{(\ell)} = P_{\theta'}^{(\ell)} \Rightarrow g(\theta) = g(\theta')$ holds.

Proposition 2 (The target is more identifiable than the latent model)**.** *If $\mathbb{E}_\theta X_t^2 < \infty$, then G_N is identified by $P_\theta^{(N)}$, irrespective of whether the latent parametrization is identified. If $\sum_{h\in\mathbb{Z}} |\operatorname{Cov}(X_0, X_h)| < \infty$, then v_∞ is identified by the full stationary observed law. In general, no fixed finite block identifies v_∞ without structural or covariance tail restrictions.*

Proof. The first assertion follows from

$$G_N = \int \left(\sum_{t=1}^{N} x_t \right)^2 dP^{(N)} - \left\{ \int \sum_{t=1}^{N} x_t \, dP^{(N)} \right\}^2 .$$

The second follows from $v_\infty = \gamma_0 + 2\sum_{h\geq 1} \gamma_h$, where every fixed-lag covariance is a moment of the observed law. A finite block contains no information about unrestricted covariances beyond its length. □

3.1 Markov submodel

Let

$$P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}, \qquad \lambda = 1 - a - b.$$

For a known number of states, a full-rank transition matrix ($\lambda \neq 0$) and linearly independent emission laws, the nonparametric two-state HMM is identified from the law of three consecutive observations, up to label permutation [3]. For two probability measures, distinctness implies linear independence; hence $\mu_1 \neq \mu_2$ is sufficient, although not necessary, because it implies $Q_1 \neq Q_2$. The convention $\mu_1 < \mu_2$ fixes the label.

The assumptions are substantive. If $Q_1 = Q_2$, latent dynamics are invisible. If $\lambda = 0$, the latent states are independent over time and the observations reduce to an ordinary mixture; an unrestricted nonparametric mixture decomposition need not be identified. Conversely, $\mu_1 = \mu_2$ does not alone destroy identification if $Q_1 \neq Q_2$, but it does remove the state signal from ordinary autocovariances of X_t.

3.2 Semi-Markov model

We assume that the embedded chain alternates, so that successive sojourns are in different states. If embedded self-transitions are allowed, a single calendar-time run in state i can be split into an arbitrary number of consecutive sojourns, and F_i is not identified even if the complete latent path S were observed.

The Markov identification theorem cannot simply be transferred to an HSMM. With duration bounds $D_i^{\max} < \infty$, the augmented state $E_t = (S_t, R_t)$ is a finite Markov chain, but all augmented states sharing the same value of S_t also share the same emission law. Thus the emission family in the expanded HMM contains repeated measures and violates the linear-independence hypothesis of the preceding theorem. A dedicated injectivity argument is required.

Moreover, an arbitrary infinite-support F_i is not identified by any fixed finite block. For a block of length ℓ, probability mass can be redistributed among at least three duration points greater than ℓ while preserving total tail mass and its first moment. This leaves all latent block probabilities of length ℓ unchanged, but can change the second duration moment. Consequently, from $P_\theta^{(\ell)}$ alone one cannot generally identify F_i or even ν_i^2. Autocovariances alone are weaker still and need not separate F_1 from F_2.

For finite-sample HSMM inference we therefore impose one of the following verifiable model restrictions:

1. F_i belongs to a fixed finite-dimensional family $F_i(\cdot;\eta_i)$ and a finite observed-block moment map is injective up to labels; or

2. F_i has known finite support and the structured HSMM likelihood is injective up to labels; or

3. a sieve $F_i^{(D)}$ is used and its truncation bias is explicitly retained.

We write the quantitative local form of this requirement as

$$\inf_{\vartheta\in U(\theta_0)} s_{\min}\{DM_H(\vartheta)\} \geq \kappa_{\mathrm{id}} > 0, \tag{3}$$

where $M_H(\theta)$ is the selected vector of observable moments. This is an assumption to be verified for the chosen duration and emission families, not a consequence of $\mu_1 \neq \mu_2$ alone.

There is nevertheless a useful constructive sufficient condition. Assume that the two-component marginal mixture is identifiable within the selected emission class:

$$\pi_1 G_1 + \pi_2 G_2 = \pi_1' G_1' + \pi_2' G_2' \quad \Longrightarrow \quad (\pi_i', G_i')_{i=1}^2 = (\pi_{s(i)}, G_{s(i)})_{i=1}^2 \tag{4}$$

for some label permutation s. This holds, for example, for many regular finite-dimensional mixture families, but not for unrestricted probability measures.

Proposition 3 (Constructive HSMM identification under mixture identifiability)**.** *Assume* (4)*, $G_1 \neq G_2$, finite mean durations, stationary initialization, and strict alternation. Then the complete observed law identifies (G_1, G_2, F_1, F_2) up to label permutation. If* $\mathrm{supp}(F_i) \subseteq \{1, \ldots, D\}$*, the block law of length $D+2$ is sufficient, though not necessarily minimal.*

Proof. The marginal law identifies (π_i, G_i) by (4). Since two distinct probability measures are linearly independent, their m-fold tensor products are linearly independent; hence the mixture decomposition of $\mathcal{L}(X_{1:m})$ identifies every latent word probability $\mathbb{P}(S_{1:m} = s)$. Stationarity and alternation give

$$c := \mathbb{P}(S_0 = 1, S_1 = 2) = \mathbb{P}(S_0 = 2, S_1 = 1) = \frac{1}{\tau_1 + \tau_2}, \qquad \tau_i = \frac{\pi_i}{c}.$$

For $j \neq i$ and $d \geq 1$,

$$f_i(d) = \frac{\mathbb{P}(S_0 = j, S_1 = \cdots = S_d = i, S_{d+1} = j)}{c}.$$

All duration masses are therefore recovered from the complete law; a support bound D makes blocks through length $D+2$ sufficient. ☐

Corollary 4 (Gaussian emission specialization)**.** *If $G_i = \mathcal{N}(\mu_i, \sigma_i^2)$, $\pi_i \in (0,1)$, and $(\mu_1, \sigma_1^2) \neq (\mu_2, \sigma_2^2)$, finite Gaussian-mixture identifiability [2] verifies* (4)*. Hence the complete observed law identifies*

$$(\mu_i, \sigma_i^2, F_i, \tau_i, \nu_i^2)_{i=1}^2$$

up to labels whenever the displayed duration moments exist. Under $\mathrm{supp}(F_i) \subseteq \{1, \ldots, D\}$*, the block law through length $D+2$ identifies the same tuple. The restriction $\mu_1 < \mu_2$ fixes labels when the means are separated; if the means coincide but the variances differ, another deterministic labeling convention is required.*

The separation condition $\mu_1 \neq \mu_2$ alone is not a replacement for (4). At the i.i.d. latent boundary, different decompositions of the same one-dimensional mixture can yield different emission and geometric run-length parameters while producing the same i.i.d. observed process. Thus neither separated means nor nondegenerate sojourn laws suffice on an unrestricted emission class. If $G_1 = G_2$, all latent dynamics are invisible.

4 Estimators

4.1 A model-free benchmark

Let $\widehat{\gamma}_h$ be an empirical autocovariance. For fixed $N < T$,

$$\widehat{G}_N^{\mathrm{dir}} = N\widehat{\gamma}_0 + 2\sum_{h=1}^{N-1}(N-h)\widehat{\gamma}_h \tag{5}$$

estimates G_N without identifying any latent parameter. Deterministically,

$$|\widehat{G}_N^{\mathrm{dir}} - G_N| \le N^2 \max_{0\le h<N} |\widehat{\gamma}_h - \gamma_h|. \tag{6}$$

This is an essential baseline, but its use of N lags makes it inefficient when N grows with T. For the long-run target we use the positive semi-definite Bartlett HAC estimator of Newey and West [1], not the rectangularly truncated covariance sum. With $Y_t = X_t - \bar{X}_T$ and

$$\widetilde{\gamma}_h = T^{-1}\sum_{t=1}^{T-h} Y_t Y_{t+h}, \qquad w_{h,H} = 1 - \frac{h}{H+1},$$

define

$$\widehat{v}_H^{\mathrm{B}} = \widetilde{\gamma}_0 + 2\sum_{h=1}^{H} w_{h,H}\widetilde{\gamma}_h. \tag{7}$$

Extending Y_t by zero outside $\{1,\dots,T\}$ gives the exact identity

$$\widehat{v}_H^{\mathrm{B}} = \frac{1}{T(H+1)}\sum_{j=1}^{T+H}\left(\sum_{r=0}^{H} Y_{j-r}\right)^2 \ge 0. \tag{8}$$

Thus the scalar estimator is nonnegative in every sample. In contrast, rectangular weights $w_h = 1$ do not define a positive-semidefinite lag window and can produce a negative estimate of a variance. The stochastic error in (7) is of order H times the maximal covariance error, plus the covariance-tail and Bartlett-window biases.

Autocovariances alone do not generally identify the full HSMM tuple. In particular, γ_0 contains only the weighted combination $\pi_1\sigma_1^2 + \pi_2\sigma_2^2$, finitely many lags cannot identify an infinite duration tail, and the scale $(\mu_1 - \mu_2)^2$ can be confounded with the latent indicator covariance. A covariance-based GMM therefore requires injectivity of its chosen finite-dimensional moment map; higher-order moments or likelihood information may be necessary.

4.2 A stable three-moment estimator for the Markov submodel

Put

$$A = \pi_1\pi_2(\mu_1 - \mu_2)^2, \qquad B = \pi_1\sigma_1^2 + \pi_2\sigma_2^2.$$

Then

$$\gamma_0 = A + B, \qquad \gamma_h = A\lambda^h \quad (h \ge 1).$$

Fix $\rho < 1$ and define

$$\mathcal{Z}_\rho = \{(A,B,\lambda) : A, B \ge 0,\ |\lambda| \le \rho\}, \qquad M(A,B,\lambda) = (A+B, A\lambda, A\lambda^2).$$

Rather than form the unstable ratio $\widehat{\gamma}_2/\widehat{\gamma}_1$, let

$$\widehat{\zeta} \in \arg\min_{\zeta\in\mathcal{Z}_\rho} \|M(\zeta) - (\widehat{\gamma}_0, \widehat{\gamma}_1, \widehat{\gamma}_2)\|_\infty, \qquad \widehat{G}_N^{\mathrm{M}} = G_N(\widehat{\zeta}). \tag{9}$$

Theorem 5 (Functional stability without a signal lower bound)**.** *Suppose $\zeta, \zeta' \in \mathcal{Z}_\rho$ and set $e_j = |M_j(\zeta') - M_j(\zeta)|$. Define*

$$S_{0,N}(\rho) = \sum_{k=0}^{N-2} (N-1-k)\rho^k,$$

$$S_{1,N}(\rho) = \sum_{k=1}^{N-2} k(N-1-k)\rho^{k-1}.$$

For $0 \le \rho < 1$ these coefficients contain no unresolved summation:

$$S_{0,N}(\rho) = \frac{(N-1) - N\rho + \rho^N}{(1-\rho)^2}, \tag{10}$$

$$S_{1,N}(\rho) = \frac{(N-2) - N\rho + N\rho^{N-1} - (N-2)\rho^N}{(1-\rho)^3}. \tag{11}$$

Then

$$|G_N(\zeta') - G_N(\zeta)| \le Ne_0 + 2S_{0,N}(\rho)e_1 + 2S_{1,N}(\rho)(\rho e_1 + e_2), \tag{12}$$

$$|v_\infty(\zeta') - v_\infty(\zeta)| \le e_0 + \frac{2e_1}{1-\rho} + \frac{2(\rho e_1 + e_2)}{(1-\rho)^2}. \tag{13}$$

In particular, no lower bound on A or $|\lambda|$ is required.

Proof. Write $u = A\lambda = \gamma_1$ and $v = A\lambda^2 = \gamma_2$. For $k \ge 1$,

$$|u\lambda^k - u'\lambda'^k| \le \rho^k e_1 + k\rho^{k-1}(\rho e_1 + e_2),$$

because $|u'(\lambda - \lambda')| \le \rho|u - u'| + |u\lambda - u'\lambda'|$. Insert this inequality into $G_N = N\gamma_0 + 2\sum_{k=0}^{N-2}(N-1-k)u\lambda^k$. Letting $N \to \infty$ after normalization gives (13); equivalently, sum the two geometric series. $\square$

The theorem is deliberately functional-specific. When $A = 0$ or $\lambda = 0$, the latent decomposition is singular but every positive-lag covariance vanishes, so the target remains stable. This avoids an artificial factor $1/|\gamma_1|$ introduced by ratio inversion.

4.3 HSMM minimum distance and likelihood

For a parametric or finite-support HSMM let

$$\widehat{M}_H = (\widehat{m}_1, \ldots, \widehat{m}_d), \qquad M_H(\theta) = \mathbb{E}_\theta \widehat{M}_H,$$

where the moments may include marginal moments, autocovariances, and, when needed for identification, third-order lagged moments. Define

$$\widehat{\theta}_{\mathrm{MD}} \in \arg\min_{\vartheta \in \Theta} \|W^{1/2}\{\widehat{M}_H - M_H(\vartheta)\}\|_2^2, \quad \widehat{G}_N = G_N(\widehat{\theta}_{\mathrm{MD}}), \quad \widehat{v}_\infty = v_\infty(\widehat{\theta}_{\mathrm{MD}}). \tag{14}$$

The exact $G_N(\widehat{\theta})$ is evaluated by the coefficient-extraction or renewal recursion of Lemma 1.

An alternative is the stationary HSMM maximum-likelihood estimator. For bounded durations, augment by residual time $E_t = (i, r)$. Its transition matrix has

$$Q_F((i,r),(i,r-1)) = 1 \quad (r > 1), \qquad Q_F((i,1),(3-i,d)) = f_{3-i}(d),$$

and stationary initial law

$$\alpha_{i,r} = \mathbb{P}(S_1 = i, R_i = r) = \frac{\mathbb{P}(D_i \geq r)}{\tau_1 + \tau_2}. \tag{15}$$

Thus the first segment must be treated as left-censored through the equilibrium residual law; the final segment is right-censored. Standard forward–backward recursions provide posterior state and segment counts. The Gaussian-emission M-step is a posterior-weighted update of μ_i and σ_i^2, while duration probabilities are updated from posterior segment counts with boundary censoring included in the likelihood.

For theory, $\widehat{\theta}$ should mean a global constrained likelihood or minimum-distance minimizer. EM is a numerical method for locating such a point; without an initialization-basin argument it does not itself supply a finite-sample guarantee.

5 Non-asymptotic propagation

Theorem 6 (Coordinatewise non-asymptotic delta method)**.** *Let $g : \Theta \to \mathbb{R}$ be continuously differentiable on a convex neighborhood U containing θ and $\widehat{\theta}$ on an event E. Put*

$$L_j = \sup_{\vartheta \in U} |\partial_j g(\vartheta)|.$$

If

$$\mathbb{P}(|\widehat{\theta}_j - \theta_j| > t) \leq \delta_j(t, T),$$

then

$$\mathbb{P}(|g(\widehat{\theta}) - g(\theta)| > \epsilon) \leq \mathbb{P}(E^c) + \inf_{\substack{t_j > 0 \\ \sum_j L_j t_j \leq \epsilon}} \sum_j \delta_j(t_j, T). \tag{16}$$

If, for all $x > 0$, simultaneously with probability at least $1 - 2de^{-x}$,

$$|\widehat{\theta}_j - \theta_j| \leq a_j \sqrt{x/T_{\mathrm{eff}}} + b_j x/T_{\mathrm{eff}},$$

then on E

$$|g(\widehat{\theta}) - g(\theta)| \leq \sum_j L_j \left(a_j \sqrt{x/T_{\mathrm{eff}}} + b_j x/T_{\mathrm{eff}} \right). \tag{17}$$

Proof. On the intersection of the coordinate events, the multivariate mean-value formula gives $|g(\widehat{\theta}) - g(\theta)| \leq \sum_j L_j t_j$. The union bound proves (16); substituting the displayed Bernstein radii proves (17). □

5.1 Observable-moment concentration and the Markov corollary

The concentration step can be made explicit by Markovizing the semi-Markov process. Put $Z_t = (S_t, R_t, X_t)$ and, for a maximum lag H, put $W_t^{(H)} = (Z_t, \ldots, Z_{t+H})$. Under bounded duration support, $Z = (Z_t)$ and each window process $W^{(H)}$ are stationary Markov chains. Let $g_H > 0$ denote the pseudo-spectral gap of $W^{(H)}$ (or any certified lower bound for it). This formulation is useful because the lag product is the bounded additive functional

$$f_h(W_t^{(H)}) = (X_t - m_X)(X_{t+h} - m_X), \qquad m_X = \mathbb{E}X_t.$$

Thus concentration of empirical autocovariances is an application of an existing Markov-chain additive-functional inequality, rather than a new semi-Markov concentration principle.

The applications below use the corrected author version of Paulin [12, Assumption 3.1 and Theorem 3.11, equation (3.25)]. Accordingly, each window chain is assumed time-homogeneous, ϕ-irreducible, and aperiodic on a Polish state space, with a unique stationary law; the equilibrium initialization makes the observed window stationary. For a bounded measurable f with $|f| \le K$, one has $V_f := \mathrm{Var}_\pi(f) \le K^2$ and $|f - \pi f| \le 2K$. Solving Paulin's quadratic exponent for the empirical-average deviation gives, for $m \ge 1$, $g > 0$, $K > 0$, and $y > 0$,

$$\mathfrak{b}_{\mathrm{P}}(m, g, K, y) = \frac{K}{mg}\left[20y + \sqrt{400y^2 + 8yg\{m + g^{-1}\}}\right], \tag{18}$$

and hence

$$\mathbb{P}_\pi\left\{\left|m^{-1}\sum_{t=1}^{m} f(W_t) - \pi f\right| > \mathfrak{b}_{\mathrm{P}}(m, g, K, y)\right\} \le 2e^{-y}.$$

No unspecified Bernstein constant remains in this radius.

Proposition 7 (Empirical autocovariances from a finite-support HSMM)**.** *Assume $|X_t - m_X| \le R$ almost surely and $T > H$. Define the common-sample autocovariances*

$$\widetilde{\gamma}_h = \frac{1}{T-H}\sum_{t=1}^{T-H}(X_t - m_X)(X_{t+h} - m_X), \qquad 0 \le h \le H,$$

and let $\widehat{\gamma}_h$ replace m_X by $\overline{X}_T = T^{-1}\sum_{t=1}^{T} X_t$. Set

$$b_\mu(x) = \mathfrak{b}_{\mathrm{P}}(T, g_0, R, x + \log 4), \tag{19}$$

$$b_{\gamma,H}(x) = \mathfrak{b}_{\mathrm{P}}(T - H, g_H, R^2, x + \log\{2(H+1)\}), \tag{20}$$

$$r_{T,H}(x) = b_{\gamma,H}(x) + 2Rb_\mu(x) + b_\mu(x)^2. \tag{21}$$

Then, for every $x > 0$,

$$\mathbb{P}\left\{\max_{0\le h\le H}|\widehat{\gamma}_h - \gamma_h| > r_{T,H}(x)\right\} \le 4e^{-x}. \tag{22}$$

Proof. For each h, apply the chosen bounded additive-functional Bernstein inequality to $f_h(W_t^{(H)})$ and take a union bound over $h = 0, \ldots, H$. This gives the term $b_{\gamma,H}(x)$. Apply the same inequality to $X_t - m_X$ to obtain $|\overline{X}_T - m_X| \le b_\mu(x)$. Finally, with $d = \overline{X}_T - m_X$ and $Y_t = X_t - m_X$,

$$|(Y_t - d)(Y_{t+h} - d) - Y_t Y_{t+h}| \le 2R|d| + d^2.$$

A second union bound proves (22). □

For Gaussian or other sub-Gaussian emissions, one may instead apply the unbounded-function renewal inequality of [13, 14] to the sub-exponential products, or use clipped observations and add the analytically controlled clipping bias. Finite emission variance by itself is not sufficient for (22).

Corollary 8 (Direct finite-horizon HSMM variance bound)**.** *Let $H = N - 1 < T$ and define $\widehat{G}_N^{\mathrm{dir}}$ by* (5)*, using the common-sample autocovariances of Proposition 7. Then*

$$\mathbb{P}\left\{|\widehat{G}_N^{\mathrm{dir}} - G_N| > N^2 r_{T,N-1}(x)\right\} \le 4e^{-x}. \tag{23}$$

Proof. On the event in Proposition 7,

$$|\widehat{G}_N^{\mathrm{dir}} - G_N| \le \left\{N + 2\sum_{h=1}^{N-1}(N-h)\right\} r_{T,N-1}(x) = N^2 r_{T,N-1}(x).$$

□

Corollary 9 (Finite-sample Markov functional bound). *Let $r_T(x) = r_{T,2}(x)$ denote the radius in* (21). *For the estimator* (9)*, with the convention that any minimizer may be selected,*

$$\mathbb{P}\{|\widehat{G}_N^{\mathrm{M}} - G_N| > 2L_N^{\mathrm{mom}}(\rho) r_T(x)\} \le 4e^{-x}, \tag{24}$$

where

$$L_N^{\mathrm{mom}}(\rho) = N + 2S_{0,N}(\rho) + 2(1+\rho)S_{1,N}(\rho).$$

The analogous bound for v_∞ replaces this constant by

$$L_\infty^{\mathrm{mom}}(\rho) = 1 + \frac{2}{1-\rho} + \frac{2(1+\rho)}{(1-\rho)^2}.$$

Proof. The true moment vector is feasible, so minimum-distance optimality and the triangle inequality give $\|M(\widehat{\zeta}) - M(\zeta)\|_\infty \le 2r_T(x)$. Apply Theorem 5 coordinatewise. □

Equivalently, the requested ϵ–δ form is the computable bound

$$\begin{aligned} &\mathbb{P}\{|\widehat{G}_N^{\mathrm{M}} - G_N| > \epsilon\} \le \delta_{N,T}^{\mathrm{M}}(\epsilon), \\ &\qquad \delta_{N,T}^{\mathrm{M}}(\epsilon) = \min[1,\, 4\exp\{-\sup\{x > 0 : 2L_N^{\mathrm{mom}}(\rho) r_{T,2}(x) \le \epsilon\}\}]\,. \end{aligned} \tag{25}$$

The right-hand side is one-dimensional and monotone, so it is evaluated without simulation; throughout these inverse-radius formulas $\sup\varnothing$ is defined as 0, yielding the trivial bound 1 when ϵ is below the smallest certified radius.

For fixed $\rho < 1$, $L_N^{\mathrm{mom}}(\rho) = O(N)$ and $L_\infty^{\mathrm{mom}}(\rho) = O(1)$. Consequently,

$$\widehat{v}_\infty - v_\infty = O_{\mathbb{P}}(T^{-1/2}), \qquad \widehat{G}_N - G_N = O_{\mathbb{P}}(NT^{-1/2}),$$

under fixed mixing constants. If $N = T$, the absolute error is therefore $O_{\mathbb{P}}(\sqrt{T})$, while the normalized error remains $O_{\mathbb{P}}(T^{-1/2})$. An $O(T^{-1/2})$ claim for the *absolute* error of G_T would have the wrong scale.

There are two distinct ill-conditioned boundaries. As $|\lambda| \uparrow 1$, mixing slows, with an effective size of order $T(1-|\lambda|)$, and the functional sensitivity contributes factors of order $(1-\lambda)^{-2}$ for positive persistence. A conservative moment argument gives

$$|\widehat{v}_\infty - v_\infty| = O_{\mathbb{P}}\{T^{-1/2}(1-\lambda)^{-5/2}\}.$$

As $\lambda \to 0$, by contrast, the full latent model approaches its non-identifiable i.i.d. boundary. Parameter recovery deteriorates, but the functional-specific estimator above need not, because the serial covariance itself vanishes.

5.2 A concrete end-to-end HSMM theorem

Fix a duration bound D and a compact parameter class Θ_D of stationary, strictly alternating, two-state HSMMs with $\mathrm{supp}(F_i) \subseteq \{1,\ldots,D\}$. Fix labels, for example by $\mu_1 < \mu_2$, and exclude zero duration masses required for irreducibility. Let $\ell \ge D+2$ and choose a finite vector of bounded block features

$$\Phi = (\phi_1,\ldots,\phi_d) : \mathbb{R}^\ell \longrightarrow [-1,1]^d, \qquad \Psi(\theta) = \mathbb{E}_\theta\{\Phi(X_{1:\ell})\}.$$

From one trajectory, set $m = T - \ell + 1$ and

$$\widehat{\Psi}_T = \frac{1}{m}\sum_{t=1}^{m} \Phi(X_{t:t+\ell-1}), \qquad \widehat{\theta}_T \in \arg\min_{\vartheta\in\Theta_D} \|\widehat{\Psi}_T - \Psi(\vartheta)\|_\infty. \tag{26}$$

This is a fully specified global minimum-distance estimator; unlike EM, its definition does not depend on a local numerical optimum.

For any estimand $g : \Theta_D \to \mathbb{R}$, define its observable identification modulus

$$\omega_{g,\Theta_D}(u) = \sup\{|g(\theta) - g(\vartheta)| : \theta, \vartheta \in \Theta_D,\ \|\Psi(\theta) - \Psi(\vartheta)\|_\infty \le u\}. \tag{27}$$

This quantity is finite by compactness and is numerically certifiable by finite-dimensional global optimization. It also records the correct impossibility statement: if $\omega_{g,\Theta_D}(u)$ does not tend to zero as $u \downarrow 0$, then the functional is not uniformly identifiable from the selected moments.

Theorem 10 (End-to-end block-moment bound for a bounded-duration HSMM)**.** *Assume that the stationary augmented window chain of length ℓ has pseudo-spectral gap at least $\underline{g}_\ell > 0$ uniformly on Θ_D. For $x > 0$, let*

$$a_{T,\ell,d}(x) = \mathfrak{b}_{\mathrm{P}}\Big(T - \ell + 1, \underline{g}_\ell, 1, x + \log(2d)\Big). \tag{28}$$

Then, simultaneously for $g = G_N$ and $g = v_\infty$,

$$\mathbb{P}_{\theta_0}\Big[|g(\widehat{\theta}_T) - g(\theta_0)| > \omega_{g,\Theta_D}\{2a_{T,\ell,d}(x)\}\Big] \le 2e^{-x}. \tag{29}$$

If, whenever $\|\Psi(\theta) - \Psi(\vartheta)\|_\infty \le r_0$,

$$\|\theta - \vartheta\|_2 \le \kappa_{\mathrm{id}}^{-1}\|\Psi(\theta) - \Psi(\vartheta)\|_\infty \tag{30}$$

and g is L_g-Lipschitz on Θ_D, then, provided $2a_{T,\ell,d}(x) \le r_0$,

$$\mathbb{P}_{\theta_0}\Big\{|g(\widehat{\theta}_T) - g(\theta_0)| > 2L_g\kappa_{\mathrm{id}}^{-1}a_{T,\ell,d}(x)\Big\} \le 2e^{-x}. \tag{31}$$

Proof. Each coordinate of $\widehat{\Psi}_T$ is a bounded additive functional of the length-ℓ augmented window chain. The Bernstein inequality used in (18), followed by a union bound over d coordinates, gives

$$\|\widehat{\Psi}_T - \Psi(\theta_0)\|_\infty \le a_{T,\ell,d}(x)$$

outside an event of probability at most $2e^{-x}$. Since θ_0 is feasible in (26), minimum-distance optimality and the triangle inequality imply

$$\|\Psi(\widehat{\theta}_T) - \Psi(\theta_0)\|_\infty \le 2a_{T,\ell,d}(x).$$

The definition (27) proves (29). Under (30), the preceding display gives $\|\widehat{\theta}_T - \theta_0\|_2 \le 2\kappa_{\mathrm{id}}^{-1}a_{T,\ell,d}(x)$; Lipschitz continuity of g then proves (31). □

In the same notation, Theorem 10 is exactly

$$\mathbb{P}_{\theta_0}\{|g(\widehat{\theta}_T) - g(\theta_0)| > \epsilon\} \le \delta_{g,T}^{\mathrm{HSMM}}(\epsilon), \tag{32}$$

where

$$\delta_{g,T}^{\mathrm{HSMM}}(\epsilon) = \min[1,\ 2\exp\{-\sup\{x > 0 : \omega_{g,\Theta_D}\{2a_{T,\ell,d}(x)\} \le \epsilon\}\}].$$

Under (30), one may replace the modulus inside the supremum by $2L_g\kappa_{\mathrm{id}}^{-1}a_{T,\ell,d}(x)$.

The theorem gives the requested deterministic observable-moment step as the pathwise implication

$$|g(\widehat{\theta}_T) - g(\theta_0)| \le \omega_{g,\Theta_D}\Big(2\|\widehat{\Psi}_T - \Psi(\theta_0)\|_\infty\Big), \tag{33}$$

not merely as a proposed future calculation.

For Gaussian emissions a concrete bounded choice is a finite collection of Fourier cylinder features

$$\phi_{u,c}(x_{1:\ell}) = \cos(u^\top x_{1:\ell}), \qquad \phi_{u,s}(x_{1:\ell}) = \sin(u^\top x_{1:\ell}).$$

On a compact class with separated component parameters, the frequency grid must be selected so that Ψ is injective and (30) is certified. The latter can be audited by a quantitative inverse-function argument using both a lower bound on the smallest singular value of $D\Psi$ and a Lipschitz bound for $D\Psi$ inside local neighborhoods, together with a global separation check outside those neighborhoods. A singular-value bound alone is not a global inverse theorem. This audit is essential: mean separation alone does not imply a uniform $\kappa_{\mathrm{id}} > 0$.

For finite duration support, the augmented transition matrix should also satisfy a quantitative geometric-ergodicity bound

$$\|(Q_F - \Pi_F)^h\| \le C_{\mathrm{mix}}\rho_F^h, \qquad \rho_F < 1, \tag{34}$$

uniformly over the parameter class; it yields a computable lower bound for $\underline{g}_\ell$. Aperiodicity alone is qualitative and does not provide a uniform constant in (34).

6 An explicit capped-geometric Gaussian submodel

We now remove both abstract constants in Theorem 10 for one named family. Fix $D \ge 2$, known $\sigma > 0$, and

$$\theta = (a, p_1, p_2) \in \Theta_{\mathrm{cg}} := [a_-, a_+] \times [p_-, p_+]^2, \quad 0 < a_- < a_+, \quad 0 < p_- < p_+ < 1.$$

Let $D_i = \min(G_i, D)$ with $G_i \sim \mathrm{Geom}(p_i)$, so that

$$f_i(d) = p_i(1-p_i)^{d-1} \quad (d < D), \qquad f_i(D) = (1-p_i)^{D-1}.$$

This is capping, not the geometric law conditioned on $G_i \le D$. Put $Y_t = 2\mathbf{1}_{\{S_t=2\}} - 1$ and

$$X_t = aY_t + \sigma Z_t, \qquad Z_t \overset{\mathrm{iid}}{\sim} N(0,1).$$

Choose ω such that $0 < \omega a_+ < \pi/2$, define $H(x) = 2\mathbf{1}_{\{x>0\}} - 1$, and use the fixed bounded feature vector

$$\phi(x_0, x_1, x_2) = \big(\cos(\omega x_0), H(x_0), H(x_0)H(x_1), H(x_0)H(x_2), H(x_0)H(x_1)H(x_2)\big). \tag{35}$$

Let $z = (z_0, \ldots, z_4) = \mathbb{E}_\theta \phi(X_0, X_1, X_2)$ and $\delta(a) = 2\Phi_{\mathrm{N}}(a/\sigma) - 1$. Then the exact inverse is

$$\begin{aligned}
a &= \omega^{-1} \arccos\{e^{\sigma^2\omega^2/2} z_0\}, & \alpha &:= \mathbb{E}Y_t = z_1/\delta(a), \\
r_1 &:= \mathbb{E}(Y_tY_{t+1}) = z_2/\delta(a)^2, & r_2 &:= \mathbb{E}(Y_tY_{t+2}) = z_3/\delta(a)^2, \\
t_3 &:= \mathbb{E}(Y_tY_{t+1}Y_{t+2}) = z_4/\delta(a)^3, & & \\
p_1 &= \frac{1+\alpha-2r_1+r_2-t_3}{2(1-r_1)}, & p_2 &= \frac{1-\alpha-2r_1+r_2+t_3}{2(1-r_1)}.
\end{aligned} \tag{36}$$

Indeed, the directed boundary rate is $c_\theta = (1-r_1)/4 = (\tau_1+\tau_2)^{-1}$, while $\mathbb{P}(+,-,+) = c_\theta p_1$ and $\mathbb{P}(-,+,-) = c_\theta p_2$; expansion of the binary indicators gives (36).

To state a global inverse margin, define

$$\begin{aligned}
\bar\tau_D &= \frac{1-(1-p_-)^D}{p_-}, & \delta_- &= 2\Phi_{\mathrm{N}}(a_-/\sigma)-1,\\
A_a &= \frac{e^{\sigma^2\omega^2/2}}{\omega\sin(\omega a_-)}, & B_\delta &= \frac{\sqrt{2/\pi}}{\sigma},\\
L_k &= \delta_-^{-k} + kB_\delta A_a \delta_-^{-(k+1)}, \quad k=1,2,3, & K_p &= \frac{\bar\tau_D}{4}(L_1+3L_2+L_3),\\
K_{\mathrm{id,cg}} &= \sqrt{A_a^2+2K_p^2}, & \underline{\kappa}_{\mathrm{id,cg}} &= K_{\mathrm{id,cg}}^{-1}. \qquad (37)
\end{aligned}$$

Proposition 11 (Calculated identification margin)**.** *For all $\theta,\vartheta\in\Theta_{\mathrm{cg}}$,*

$$\|\theta-\vartheta\|_2 \le K_{\mathrm{id,cg}}\|\mathbb{E}_\theta\phi-\mathbb{E}_\vartheta\phi\|_\infty. \qquad (38)$$

Thus κ_{id} in (30) *may be set equal to the explicit value $\underline{\kappa}_{\mathrm{id,cg}}$ for this feature map.*

Proof. The derivative of the first coordinate of (36) is bounded by A_a. Since $|\delta'(a)|\le B_\delta$, a ratio of an observed moment by $\delta(a)^k$ changes by at most L_k times the sup-norm error in z. Moreover, $1-r_1=4c_\theta\ge 2/\bar\tau_D$. The absolute derivatives of p_i with respect to α, r_2, t_3 are at most $\bar\tau_D/4$, and $|\partial_{r_1}p_i| = |p_i-1|/(1-r_1)\le\bar\tau_D/2$. Hence $|p_i-p_i'|\le K_p\|z-z'\|_\infty$. Combining the three coordinate bounds in Euclidean norm proves (38). □

The constant in (37) is a certified global margin, not a claim that every triangle inequality is sharp. At a specified parameter, the exact local margin is

$$\kappa^{\mathrm{loc}}_{\mathrm{id,cg}}(\theta) = \|D\mathcal{I}(\Psi(\theta))\|^{-1}_{\infty\to 2} = \left\{\max_{s\in\{-1,1\}^5}\|D\mathcal{I}(\Psi(\theta))s\|_2\right\}^{-1},$$

where $\mathcal{I}$ is the inverse in (36). Its full Jacobian is displayed in the accompanying derivation.

For the mixing constant, augment by the current age: $Z_t=(S_t,A_t)\in\{1,2\}\times\{1,\dots,D\}$. For $r<D$,

$$Q((i,r),(i,r+1)) = 1-p_i, \qquad Q((i,r),(3-i,1)) = p_i,$$

and $Q((i,D),(3-i,1))=1$. Set

$$a_* = \min\{p_-,1-p_+\}, \qquad \varepsilon_* = 2a_*^3, \qquad K_* = \left\lceil\frac{\log 2}{-\log(1-\varepsilon_*)}\right\rceil. \qquad (39)$$

From any augmented state, the path consisting of a switch, a continuation, and a switch, and the path of three consecutive switches, reach $(1,1)$ and $(2,1)$ in an order that depends on the current state. Forced exits only increase their probabilities. Writing $\nu_*=\{\delta_{(1,1)}+\delta_{(2,1)}\}/2$, therefore

$$Q^3(z,\cdot)\ge\varepsilon_*\nu_*(\cdot), \qquad \sup_{z,z'} d_{\mathrm{TV}}\{Q^t(z,\cdot),Q^t(z',\cdot)\}\le(1-\varepsilon_*)^{\lfloor t/3\rfloor}. \qquad (40)$$

For the length-three window chain, two extra shifts remove the initial coordinates. In the notation of Definition 1.4 of the corrected version of Paulin [12], the preceding pairwise total-variation bound and the definition of K_* give

$$\tau_{W^{(2)}}(1/2)\le 2+3K_*.$$

Applying the first inequality in Proposition 3.4, equation (3.9), with $\epsilon=1/2$ therefore yields

$$\gamma_{\mathrm{ps}}(W^{(2)}) \ge \frac{1-1/2}{\tau_{W^{(2)}}(1/2)} \ge \frac{1}{2(2+3K_*)}.$$

Thus the calculated uniform bound is

$$\boxed{\underline{g}_3 = \{2(2+3K_*)\}^{-1} \le \gamma_{\mathrm{ps}}(W^{(2)}).} \tag{41}$$

Pointwise sharper values follow directly from the $2D \times 2D$ matrix Q. If D_η is its stationary diagonal matrix, then

$$\gamma_{\mathrm{ps}}(Q) = \max_{k\ge 1} \frac{1 - s_2^2(D_\eta^{1/2} Q^k D_\eta^{-1/2})}{k}, \tag{42}$$

where s_2 is the second singular value. This is an exact finite-matrix calculation; (41) is the uniform analytic lower bound and is not presented as optimal.

Finally, let $M = T-2$ and $y = x + \log 5$. Every coordinate f of (35) satisfies $|f| \le 1$, and hence $|f - \mathbb{E}_\pi f| \le 2$ and $V_f \le 1$. Since the observation process is stationary, its length-three window chain is taken in its stationary distribution. Apply Theorem 3.11, equation (3.25), of the corrected version of Paulin [12] to each coordinate with $n = M = T-2$, $\gamma_{\mathrm{ps}} \ge \underline{g}_3$, $C \le 2$, and $V_f \le 1$. That equation controls the sum $S = \sum_{t=1}^M f(W_t^{(2)})$. Substituting the sum threshold $t = Mr$ and requiring the exponent to be at least y gives

$$M^2 \underline{g}_3 r^2 \ge y\{8(M + 1/\underline{g}_3) + 40Mr\}.$$

Solving this quadratic inequality for r and dividing the sum threshold by M gives

$$r_T(x) = \frac{20y + \sqrt{400y^2 + 8y\underline{g}_3(M + 1/\underline{g}_3)}}{M\underline{g}_3}. \tag{43}$$

Applying the resulting bound to each of the five coordinates and taking a union bound gives a factor 5; because $y = x + \log 5$, this gives

$$\mathbb{P}_\theta\{\|\widehat{z}_T - z\|_\infty > r_T(x)\} \le 2e^{-x}.$$

For the global minimum-distance estimator over Θ_{cg}, Proposition 11 therefore yields

$$\boxed{\mathbb{P}_\theta\{\|\widehat{\theta}_T - \theta\|_2 > 2K_{\mathrm{id,cg}} r_T(x)\} \le 2e^{-x}.} \tag{44}$$

The detailed proof, exact local Jacobian, and numerical matrix audit are in `step1_explicit_identification_mixing.tex` and `step1_explicit_constants.py`.

For reference, the previously derived alternating-renewal expression

$$v_\infty = \frac{\tau_1\sigma_1^2 + \tau_2\sigma_2^2}{T_c} + (\mu_1 - \mu_2)^2 \frac{\tau_2^2\nu_1^2 + \tau_1^2\nu_2^2}{T_c^3}, \qquad T_c = \tau_1 + \tau_2, \tag{45}$$

has fully explicit sensitivities. With $\Delta = \mu_1 - \mu_2$ and $A_c = \tau_2^2\nu_1^2 + \tau_1^2\nu_2^2$,

$$\partial_{\mu_1} v_\infty = 2\Delta A_c/T_c^3, \qquad \partial_{\mu_2} v_\infty = -2\Delta A_c/T_c^3,$$

$$\partial_{\sigma_i^2} v_\infty = \pi_i, \qquad \partial_{\nu_1^2} v_\infty = \Delta^2\tau_2^2/T_c^3,$$

$$\partial_{\nu_2^2} v_\infty = \Delta^2\tau_1^2/T_c^3, \qquad \partial_{\tau_1} v_\infty = \frac{\tau_2(\sigma_1^2 - \sigma_2^2)}{T_c^2} + \Delta^2\left(\frac{2\tau_1\nu_2^2}{T_c^3} - \frac{3A_c}{T_c^4}\right),$$

$$\partial_{\tau_2} v_\infty = \frac{\tau_1(\sigma_2^2 - \sigma_1^2)}{T_c^2} + \Delta^2\left(\frac{2\tau_2\nu_1^2}{T_c^3} - \frac{3A_c}{T_c^4}\right).$$

These derivatives may be inserted directly into (16).

7 A matching local upper bound on the persistent path

Consider the symmetric Markov submodel

$$Q_u = \begin{pmatrix} 1-u & u \\ u & 1-u \end{pmatrix}, \qquad X_t = aY_t + \varepsilon_t, \qquad \varepsilon_t \overset{\text{iid}}{\sim} N(0,\sigma^2),$$

where $Y_t \in \{-1,1\}$, $a \neq 0$ and σ are fixed, and

$$v_\infty(u) = \sigma^2 + a^2 \frac{1-u}{u}.$$

This is the same one-dimensional persistence subexperiment used below for the two-point lower bound.

Fix $0 < \alpha \le 1 \le \beta$, a local scale $u_\star > 0$, and

$$\mathcal{U}_\star = [\alpha u_\star, \beta u_\star], \qquad \beta u_\star \le \frac{1}{4}. \tag{46}$$

Put

$$k = \left\lceil \frac{1}{u_\star} \right\rceil, \qquad M = \left\lfloor \frac{T-1}{k} \right\rfloor,$$

and, when $M \ge 1$, define the observable block moment

$$\widehat{R}_k = \frac{1}{a^2 M} \sum_{r=0}^{M-1} X_{1+rk} X_{1+(r+1)k}. \tag{47}$$

Since

$$R_k(u) := \mathbb{E}_u \widehat{R}_k = (1-2u)^k,$$

let

$$\mathcal{R}_\star = \left[(1-2\beta u_\star)^k, (1-2\alpha u_\star)^k \right], \qquad \widetilde{R}_k = \Pi_{\mathcal{R}_\star}(\widehat{R}_k),$$

and set

$$\widehat{u}_k = \frac{1-\widetilde{R}_k^{1/k}}{2}, \qquad \widehat{v}_{\infty,k} = \sigma^2 + a^2 \frac{1-\widehat{u}_k}{\widehat{u}_k}. \tag{48}$$

Theorem 12 (Oracle-scale local long-lag upper bound)**.** *Suppose $T-1 \ge 2k$, let $0<\delta<1$, and put $\ell_\delta = \log(6/\delta)$. Uniformly over $u \in \mathcal{U}_\star$, with probability at least $1-\delta$,*

$$|\widehat{v}_{\infty,k} - v_\infty(u)| \le \frac{e^{5\beta}}{\alpha^2} \left\{ \frac{(|a|+\sigma)^2 \sqrt{\ell_\delta}}{u_\star^{3/2} \sqrt{T-1}} + \frac{(4a^2/3 + 2\sigma^2)\ell_\delta}{u_\star^2 (T-1)} \right\}. \tag{49}$$

Proof. Write

$$Y_{t+1} = Y_t B_t, \qquad \mathbb{P}(B_t = -1) = u,$$

where the B_t's are independent. At the sampled times $t_r = 1 + rk$, the block products

$$C_r = Y_{t_r} Y_{t_{r+1}} = \prod_{j=t_r}^{t_{r+1}-1} B_j$$

are independent and satisfy

$$\mathbb{E} C_r = (1-2u)^k = R_k, \qquad \mathrm{Var}(C_r) = 1 - R_k^2.$$

Writing $\xi_r = \varepsilon_{t_r}$,

$$X_{t_r} X_{t_{r+1}} - a^2 R_k = a^2(C_r - R_k) + aY_{t_r}\xi_{r+1} + aY_{t_{r+1}}\xi_r + \xi_r\xi_{r+1}.$$

Bernstein's inequality gives, except on an event of probability $2e^{-\ell_\delta}$,

$$\left| \frac{1}{M} \sum_{r=0}^{M-1} (C_r - R_k) \right| \le \sqrt{\frac{2(1-R_k^2)\ell_\delta}{M}} + \frac{4\ell_\delta}{3M}.$$

Conditionally on Y, the sum of the two terms linear in ξ is Gaussian with variance at most $4a^2\sigma^2 M$. Hence, except on a second event of probability $2e^{-\ell_\delta}$, its absolute value is at most

$$2\sqrt{2}\,|a|\sigma\sqrt{M\ell_\delta}.$$

Finally,

$$\sum_{r=0}^{M-1} \xi_r\xi_{r+1} = \sigma^2 g^\top A_M g,$$

where $g \sim N(0, I_{M+1})$, A_M has entries $1/2$ on its two adjacent diagonals, $\|A_M\|_{\mathrm{F}}^2 = M/2$, and $\|A_M\|_{\mathrm{op}} \le 1$. The Gaussian quadratic-form bound therefore gives, except on a third event of probability $2e^{-\ell_\delta}$,

$$\left| \sum_{r=0}^{M-1} \xi_r\xi_{r+1} \right| \le \sigma^2 \left(\sqrt{2M\ell_\delta} + 2\ell_\delta \right).$$

A union bound and $1 - R_k^2 \le 1$ yield

$$|\widehat{R}_k - R_k| \le A_s \sqrt{\frac{\ell_\delta}{M}} + B_s \frac{\ell_\delta}{M}, \quad A_s = \sqrt{2}(1+s)^2, \quad B_s = \frac{4}{3} + 2s^2, \quad s = \frac{\sigma}{|a|}. \tag{50}$$

Projection onto $\mathcal{R}_\star$ cannot increase the error. Moreover,

$$(1 - 2\beta u_\star)^k \ge e^{-5\beta},$$

because $ku_\star \le 5/4$ and $\log(1-x) \ge -x/(1-x) \ge -2x$ for $0 \le x \le 1/2$. For $g_k(r) = (1 - r^{1/k})/2$, it follows that

$$\sup_{r \in \mathcal{R}_\star} |g_k'(r)| \le \frac{e^{5\beta}}{2k}.$$

Thus (50), $M \ge (T-1)/(2k)$, and $k \ge 1/u_\star$ imply

$$|\widehat{u}_k - u| \le e^{5\beta} \left\{ A_s \sqrt{\frac{u_\star \ell_\delta}{2(T-1)}} + B_s \frac{\ell_\delta}{T-1} \right\}. \tag{51}$$

Both u and $\widehat{u}_k$ lie in $\mathcal{U}_\star$, and hence

$$|\widehat{v}_{\infty,k} - v_\infty(u)| = a^2 \frac{|\widehat{u}_k - u|}{u\widehat{u}_k} \le \frac{a^2}{\alpha^2 u_\star^2} |\widehat{u}_k - u|.$$

Substitution of A_s and B_s proves (49). $\square$

The two terms in (49) have a direct regime interpretation. If $(T-1)u_\star \gg \ell_\delta$, the leading radius is

$$\frac{(|a| + \sigma)^2 \sqrt{\log(1/\delta)}}{u_\star^{3/2}\sqrt{T}}. \tag{52}$$

If $Tu_\star = O(1)$, any estimator constrained to $\mathcal{U}_\star$ obeys the deterministic diameter bound

$$|\widehat{v}_\infty - v_\infty(u)| \le \frac{a^2(\beta-\alpha)}{\alpha\beta u_\star} = O(a^2/u). \tag{53}$$

For sample sizes with $M = 0$, this bound is attained in order by taking any fixed $\widehat{u} \in \mathcal{U}_\star$. The boundary statement is therefore a saturation result, not a claim of informative relative estimation.

The choice of lag is essential. For the lag-one product $Z_t = X_t X_{t+1}$, with $\lambda = 1 - 2u$,

$$\operatorname{Var}(Z_t) + 2\operatorname{Cov}(Z_t, Z_{t+1}) = (a^2+\sigma^2)^2 - a^4\lambda^2 + 2a^2\sigma^2\lambda^2 \longrightarrow \sigma^4 + 4a^2\sigma^2. \tag{54}$$

Consequently the lag-one plug-in has order $a^2u^{-2}T^{-1/2}$ when $\sigma > 0$, worse than (52) by $u^{-1/2}$. This polynomial gap belongs to the lag-one estimator: the long-lag construction removes it. The procedure is oracle-local: both the lag $k = \lceil 1/u_\star \rceil$ and the projection interval $\mathcal{R}_\star$ require the prescribed scale $u_\star$. Theorem 12 is uniform only over the single cell $\mathcal{U}_\star$. No data-driven scale selector, and hence no adaptive bound over a union of persistence cells, is proved here.

8 Explicit two-point minimax lower bounds

Let $P_\theta^{(T)}$ denote the law of the observed trajectory in the capped-geometric Gaussian submodel of Section 6. The data length remains T and the target horizon remains N.

For fixed $p^\circ = (p_1^\circ, p_2^\circ) \in (p_-, p_+)^2$, let Q° and η° be the transition matrix and stationary law of the augmented age chain. Put

$$y(i,r) = 2\mathbf{1}_{\{i=2\}} - 1, \qquad \alpha^\circ = (\eta^\circ)^\top y, \qquad \bar{y} = y - \alpha^\circ \mathbf{1},$$
$$\Pi^\circ = \mathbf{1}(\eta^\circ)^\top, \qquad R^\circ = Q^\circ - \Pi^\circ,$$

and use the stationary inner product $\langle u, v\rangle_{\eta^\circ} = u^\top \operatorname{diag}(\eta^\circ) v$. Define

$$\begin{aligned} A_N^\circ &:= \operatorname{Var}_{p^\circ}\left(\sum_{t=1}^N Y_t\right) \\ &= N\langle \bar{y}, \bar{y}\rangle_{\eta^\circ} + 2\left\langle \bar{y},\, R^\circ\{(N-1)I - NR^\circ + (R^\circ)^N\}(I-R^\circ)^{-2}\bar{y}\right\rangle_{\eta^\circ}. \end{aligned} \tag{55}$$

If $\tau_i^\circ = \tau_D(p_i^\circ)$ and $\nu_i^{\circ\,2} = \operatorname{Var}\{\min(G_i, D)\}$, set

$$A_\infty^\circ = 4\,\frac{(\tau_2^\circ)^2 \nu_1^{\circ\,2} + (\tau_1^\circ)^2 \nu_2^{\circ\,2}}{(\tau_1^\circ + \tau_2^\circ)^3}. \tag{56}$$

The two target functionals along the amplitude submodel are therefore

$$G_N(a, p^\circ) = N\sigma^2 + a^2 A_N^\circ, \qquad v_\infty(a, p^\circ) = \sigma^2 + a^2 A_\infty^\circ. \tag{57}$$

8.1 An explicit relative-entropy envelope

For $p \in (0,1)$, define

$$\tau_D(p) = \sum_{r=0}^{D-1}(1-p)^r, \qquad e_D(p) = \sum_{r=0}^{D-2}(1-p)^r,$$

$$c_D(p) = \sum_{r=0}^{D-1} r(1-p)^r = \frac{(1-p) - D(1-p)^D + (D-1)(1-p)^{D+1}}{p^2},$$

and

$$\operatorname{kl}(u\|v) = u\log\frac{u}{v} + (1-u)\log\frac{1-u}{1-v}.$$

Lemma 13 (Complete-data entropy envelope)**.** *For $\theta = (a, p_1, p_2)$ and $\theta' = (a', p_1', p_2')$, write $T_c(\theta) = \tau_D(p_1) + \tau_D(p_2)$. Then*

$$D_{\mathrm{KL}}\Big(P_\theta^{(T)}\Big\|P_{\theta'}^{(T)}\Big) \le \mathcal{K}_T(\theta, \theta') := \log \frac{T_c(\theta')}{T_c(\theta)} + \frac{1}{T_c(\theta)} \sum_{i=1}^{2} c_D(p_i) \log \frac{1 - p_i}{1 - p_i'} + (T-1) \sum_{i=1}^{2} \frac{e_D(p_i)}{T_c(\theta)} \mathrm{kl}(p_i \| p_i') + \frac{T(a - a')^2}{2\sigma^2}. \tag{58}$$

Proof. The stationary law of the augmented age chain is

$$\eta_\theta(i, r) = \frac{(1 - p_i)^{r-1}}{T_c(\theta)}, \qquad 1 \le r \le D.$$

Consequently,

$$D_{\mathrm{KL}}(\eta_\theta \| \eta_{\theta'}) = \log \frac{T_c(\theta')}{T_c(\theta)} + \frac{1}{T_c(\theta)} \sum_{i=1}^{2} c_D(p_i) \log \frac{1 - p_i}{1 - p_i'}.$$

At an age $r < D$, the transition row is Bernoulli with switching probability p_i; at $r = D$, the exit is forced under both parameters. Stationarity thus makes the transition contribution equal to

$$(T-1) \sum_{i=1}^{2} \frac{e_D(p_i)}{T_c(\theta)} \mathrm{kl}(p_i \| p_i').$$

The conditional Gaussian contribution is

$$\sum_{t=1}^{T} D_{\mathrm{KL}}\{N(aY_t, \sigma^2) \| N(a'Y_t, \sigma^2)\} = \frac{T(a - a')^2}{2\sigma^2}.$$

The chain rule gives equality for the complete trajectory $(Z_{1:T}, X_{1:T})$; projection onto $X_{1:T}$ proves (58). □

8.2 Finite-sample two-point bound

Theorem 14 (Explicit minimax lower bound)**.** *Fix $a_0 \in [a_-, a_+)$, $p^\circ \in (p_-, p_+)^2$, and $0 < d \le a_+ - a_0$. Let $\theta_0 = (a_0, p^\circ)$ and $\theta_1 = (a_0 + d, p^\circ)$. Every estimator based on $X_{1:T}$ satisfies*

$$\max_{j=0,1} P_{\theta_j}^{(T)} \left\{ \|\widehat{\theta}_T - \theta_j\|_2 \ge \frac{d}{2} \right\} \ge \frac{1}{4} \exp\left\{ -\frac{Td^2}{2\sigma^2} \right\}, \tag{59}$$

$$\max_{j=0,1} P_{\theta_j}^{(T)} \left\{ \left|\widehat{G}_{N,T} - G_N(\theta_j)\right| \ge \frac{A_N^\circ}{2}(2a_0 d + d^2) \right\} \ge \frac{1}{4} \exp\left\{ -\frac{Td^2}{2\sigma^2} \right\}, \tag{60}$$

$$\max_{j=0,1} P_{\theta_j}^{(T)} \left\{ |\widehat{v}_{\infty,T} - v_\infty(\theta_j)| \ge \frac{A_\infty^\circ}{2}(2a_0 d + d^2) \right\} \ge \frac{1}{4} \exp\left\{ -\frac{Td^2}{2\sigma^2} \right\}. \tag{61}$$

Proof. Lemma 13, with p° fixed, gives

$$D_{\mathrm{KL}}\Big(P_{\theta_0}^{(T)}\Big\|P_{\theta_1}^{(T)}\Big) \le \frac{Td^2}{2\sigma^2}.$$

The nearest-point test associated with any estimator and the Bretagnolle–Huber inequality [8] give

$$\max_{j=0,1} P_{\theta_j}^{(T)} \left\{ d_g(\widehat{g}_T, g(\theta_j)) \ge \frac{d_g(g(\theta_0), g(\theta_1))}{2} \right\} \ge \frac{1}{4} \exp\Big\{ -D_{\mathrm{KL}}(P_{\theta_0}^{(T)} \| P_{\theta_1}^{(T)}) \Big\}.$$

Here d_g is Euclidean distance for $g(\theta) = \theta$ and absolute distance for the scalar functionals. Equation (57) supplies the three separations in (59)–(61). □

Corollary 15 (Necessary confidence radii and matching rate). *For $0 < \delta < 1/8$, put*

$$d_{T,\delta} = \sigma\sqrt{\frac{2}{T}\log\frac{1}{8\delta}}, \qquad y_\delta = \log\frac{10}{\delta}, \qquad M = T - 2,$$

and assume $d_{T,\delta} \le a_+ - a_0$. If $\mathfrak{r}_{T,\delta}(g)$ denotes the smallest uniform $(1-\delta)$-confidence radius for g, namely

$$\mathfrak{r}_{T,\delta}(g) = \inf_{\widehat{g}} \inf \left\{ r > 0 : \sup_{\theta\in\Theta_{\mathrm{cg}}} P_\theta^{(T)}\{\ell_g(\widehat{g}, g(\theta)) > r\} \le \delta \right\},$$

where ℓ_θ is Euclidean distance and the two functional losses are absolute distance, then

$$\mathfrak{r}_{T,\delta}(\theta) \ge \frac{d_{T,\delta}}{2}, \tag{62}$$

$$\mathfrak{r}_{T,\delta}(G_N) \ge A_N^\circ \left(a_0 d_{T,\delta} + \frac{d_{T,\delta}^2}{2} \right), \tag{63}$$

$$\mathfrak{r}_{T,\delta}(v_\infty) \ge A_\infty^\circ \left(a_0 d_{T,\delta} + \frac{d_{T,\delta}^2}{2} \right). \tag{64}$$

For the estimator of Section 6, define

$$U_{T,\delta} = 2K_{\mathrm{id,cg}} \left[\frac{40 y_\delta}{M\underline{g}_3} + \sqrt{\frac{8y_\delta}{M\underline{g}_3}\left(1 + \frac{1}{M\underline{g}_3}\right)} \right].$$

Then

$$\mathfrak{r}_{T,\delta}(\theta) \le U_{T,\delta}, \qquad \mathfrak{r}_{T,\delta}(g) \le L_g U_{T,\delta}, \quad g \in \{G_N, v_\infty\}, \tag{65}$$

where L_g is the Lipschitz constant of g on Θ_{cg}. Hence, for a fixed interior parameter box,

$$\mathfrak{r}_{T,\delta}(\theta) \asymp \sqrt{\frac{\log(1/\delta)}{T}}, \qquad \mathfrak{r}_{T,\delta}(v_\infty) \asymp \sqrt{\frac{\log(1/\delta)}{T}},$$

up to structural constants. Moreover,

$$\frac{A_N^\circ}{N} \longrightarrow A_\infty^\circ > 0, \tag{66}$$

so the rate for the unnormalised target is

$$\mathfrak{r}_{T,\delta}(G_N) \asymp N\sqrt{\frac{\log(1/\delta)}{T}}$$

whenever N is in the regime where the uniform $O(N)$ Lipschitz bound applies.

Proof. With $d = d_{T,\delta}$, the right side of (59)–(61) equals 2δ, proving (62)–(64). Setting $x = \log(2/\delta)$ in the upper bound of Section 6, and using $\sqrt{u+v} \le \sqrt{u} + \sqrt{v}$, gives (65). Finally,

$$\frac{A_N^\circ}{N} = \mathrm{Var}(Y_0) + 2\sum_{h=1}^{N-1}\left(1 - \frac{h}{N}\right)\mathrm{Cov}(Y_0, Y_h) \longrightarrow A_\infty^\circ$$

by the geometric covariance bound. On the compact parameter box, the covariances and their first parameter derivatives are uniformly summable, which gives the corresponding $O(N)$ upper Lipschitz bound. □

8.3 Persistence-sensitive lower bound

On the symmetric submodel $p_1 = p_2 = p$, define

$$\omega_D(p) = \frac{e_D(p)}{\tau_D(p)}, \qquad \nu_D^2(p) = \sum_{r=0}^{D-1}(2r+1)(1-p)^r - \tau_D(p)^2,$$

$$H_D(p) = \frac{\nu_D^2(p)}{\tau_D(p)}, \qquad v_\infty(a,p,p) = \sigma^2 + a^2 H_D(p),$$

and

$$K_D^{\mathrm{age}}(p,q) = \log\frac{\tau_D(q)}{\tau_D(p)} + \frac{c_D(p)}{\tau_D(p)}\log\frac{1-p}{1-q}.$$

All derivatives in the local bound below are explicit finite sums:

$$\tau_D'(p) = -\sum_{r=1}^{D-1} r(1-p)^{r-1},$$

$$\{\nu_D^2\}'(p) = -\sum_{r=1}^{D-1} r(2r+1)(1-p)^{r-1} - 2\tau_D(p)\tau_D'(p),$$

$$H_D'(p) = \frac{\{\nu_D^2\}'(p)\tau_D(p) - \nu_D^2(p)\tau_D'(p)}{\tau_D(p)^2}.$$

Proposition 16 (Explicit persistence lower bound)**.** *For interior $p, q \in [p_-, p_+]$, every estimator satisfies*

$$\max_{\vartheta\in\{p,q\}} P_{(a,\vartheta,\vartheta)}^{(T)}\left\{|\widehat{v}_{\infty,T} - v_\infty(a,\vartheta,\vartheta)| \geq \frac{a^2}{2}|H_D(q) - H_D(p)|\right\}$$
$$\geq \frac{1}{4}\exp\left[-K_D^{\mathrm{age}}(p,q) - (T-1)\omega_D(p)\,\mathrm{kl}(p\|q)\right]. \tag{67}$$

If $q_T = p + h/\sqrt{T}$ and $H_D'(p) \neq 0$, then

$$K_D^{\mathrm{age}}(p,q_T) + (T-1)\omega_D(p)\,\mathrm{kl}(p\|q_T) \longrightarrow \frac{h^2\omega_D(p)}{2p(1-p)}, \tag{68}$$

$$\frac{\sqrt{T}}{2}|v_\infty(a,q_T,q_T) - v_\infty(a,p,p)| \longrightarrow \frac{a^2|H_D'(p)h|}{2}. \tag{69}$$

Thus the local unavoidable scale is

$$a^2|H_D'(p)|\sqrt{\frac{p(1-p)}{T\omega_D(p)}}. \tag{70}$$

Proof. Apply Lemma 13 to (a,p,p) and (a,q,q), and then apply the two-point testing inequality used in Theorem 14. Equations (68) and (69) follow from the second-order expansion

$$\mathrm{kl}(p\|p+u) = \frac{u^2}{2p(1-p)} + O(u^3)$$

and first-order differentiation of H_D. □

8.4 Role of the exact local identification modulus

Let $\mathcal{I}$ be the inverse map in (36), let $J_\theta = D\mathcal{I}(\Psi(\theta))$, and let

$$\Sigma_\Phi(\theta) = \sum_{k\in\mathbb{Z}} \mathrm{Cov}_\theta\{\phi(X_0, X_1, X_2), \phi(X_k, X_{k+1}, X_{k+2})\}.$$

Proposition 17 (Plug-in fluctuation)**.** *At an interior parameter, put*

$$\lambda_\Phi(\theta) := \lambda_{\min}\{\Sigma_\Phi(\theta)\},$$

and suppose that $\lambda_\Phi(\theta) > 0$. The direct inverse estimator $\widetilde{\theta}_T = \mathcal{I}(\widetilde{z}_T)$, locally extended outside the range of $\mathcal{I}$, satisfies

$$\sqrt{T}(\widetilde{\theta}_T - \theta) \implies N\{0, J_\theta \Sigma_\Phi(\theta) J_\theta^\top\}.$$

Writing

$$K^{\mathrm{loc}}_{\mathrm{id,cg}}(\theta) = \|J_\theta\|_{\infty\to 2} = \{\kappa^{\mathrm{loc}}_{\mathrm{id,cg}}(\theta)\}^{-1},$$

for every $c > 0$,

$$\liminf_{T\to\infty} P_\theta\left\{\sqrt{T}\|\widetilde{\theta}_T - \theta\|_2 \geq c K^{\mathrm{loc}}_{\mathrm{id,cg}}(\theta)\sqrt{\frac{\lambda_\Phi(\theta)}{5}}\right\} \geq 2\Phi_{\mathrm{N}}(-c). \tag{71}$$

Proof. Geometric ergodicity and boundedness of ϕ give the multivariate Markov-chain central limit theorem; the displayed weak limit follows by the delta method. Since

$$\|J_\theta\|_{\infty\to 2} \leq \sqrt{5}\,\|J_\theta\|_{2\to 2},$$

$$\lambda_{\max}\{J_\theta \Sigma_\Phi(\theta) J_\theta^\top\} \geq \frac{\lambda_\Phi(\theta)}{5}\{K^{\mathrm{loc}}_{\mathrm{id,cg}}(\theta)\}^2.$$

Projection onto a maximizing eigenvector gives (71). □

Remark 18 (Scope of rate matching). Corollary 15 proves minimax-rate optimality in T, $\log(1/\delta)$, and N on a fixed interior box. Proposition 17 shows how the exact local inverse modulus enters the fluctuation of the specified moment plug-in estimator. It does not make the global factor $K_{\mathrm{id,cg}}/\sqrt{\underline{g}_3}$ a sharp minimax constant. Indeed, rescaling the feature vector changes $K_{\mathrm{id,cg}}$ while leaving the observation experiment unchanged; the invariant local object is $J_\theta\Sigma_\Phi(\theta)J_\theta^\top$. Likewise, $\underline{g}_3$ is a certified uniform Doeblin lower bound, whereas (67) is governed by the explicit local information coefficient $\omega_D(p)/\{p(1-p)\}$.

8.5 Quantitative comparison with the closest minimax antecedents

Two-state HMM parameter bounds. First consider the uncapped Markov benchmark $D = \infty$, and write

$$Q = \begin{pmatrix} 1-p & p \\ q & 1-q \end{pmatrix}, \qquad s = p+q, \qquad \lambda = 1-s.$$

For $Y_t = 2\mathbf{1}_{\{S_t=2\}} - 1$, stationarity gives

$$B(p,q) := \mathrm{Var}(Y_t) = \frac{4pq}{s^2}, \qquad \mathrm{Cov}(Y_t, Y_{t+h}) = B(p,q)\lambda^h.$$

Consequently,

$$v_\infty(p,q,a) = \sigma^2 + a^2 B(p,q)\frac{1+\lambda}{1-\lambda} = \sigma^2 + \frac{4a^2pq(2-p-q)}{(p+q)^3}, \tag{72}$$

$$G_N(p,q,a) = N\sigma^2 + a^2 B(p,q) F_N(\lambda), \tag{73}$$

where

$$F_N(\lambda) = N + 2\sum_{h=1}^{N-1}(N-h)\lambda^h$$
$$= N\frac{1+\lambda}{1-\lambda} - \frac{2\lambda(1-\lambda^N)}{(1-\lambda)^2}, \tag{74}$$
$$F_N'(\lambda) = \frac{2\{(N-1)-(N+1)\lambda+(N+1)\lambda^N-(N-1)\lambda^{N+1}\}}{(1-\lambda)^3}. \tag{75}$$

The comparison with [7] can be made literal, despite their finite-alphabet assumption, by applying their estimator to the observable coarsening $U_t = \mathbf{1}_{\{X_t>0\}}$. The two Bernoulli emission vectors are

$$f_1 = \{\Phi_\mathrm{N}(a/\sigma), \Phi_\mathrm{N}(-a/\sigma)\}, \qquad f_2 = \{\Phi_\mathrm{N}(-a/\sigma), \Phi_\mathrm{N}(a/\sigma)\},$$

so their emission-separation coordinate is

$$\phi_3 = \|f_1 - f_2\|_2 = \sqrt{2}\{2\Phi_\mathrm{N}(a/\sigma) - 1\}.$$

With their remaining coordinates

$$\phi_1 = \frac{q-p}{p+q}, \qquad \phi_2 = 1-p-q = \lambda,$$

one has

$$a = a(\phi_3) = \sigma\Phi_\mathrm{N}^{-1}\left(\frac{1+\phi_3/\sqrt{2}}{2}\right), \qquad a'(\phi_3) = \frac{\sigma}{2\sqrt{2}\,\varphi_\mathrm{N}\{a(\phi_3)/\sigma\}},$$

where φ_N is the standard normal density. Hence

$$v_\infty(\phi) = \sigma^2 + a(\phi_3)^2(1-\phi_1^2)\frac{1+\phi_2}{1-\phi_2}, \tag{76}$$
$$G_N(\phi) = N\sigma^2 + a(\phi_3)^2(1-\phi_1^2)F_N(\phi_2). \tag{77}$$

The required delta map is therefore explicit:

$$\partial_{\phi_1} v_\infty = -2a^2\phi_1\frac{1+\phi_2}{1-\phi_2}, \qquad \partial_{\phi_2} v_\infty = \frac{2a^2(1-\phi_1^2)}{(1-\phi_2)^2},$$
$$\partial_{\phi_3} v_\infty = 2aa'(\phi_3)(1-\phi_1^2)\frac{1+\phi_2}{1-\phi_2}, \tag{78}$$
$$\partial_{\phi_1} G_N = -2a^2\phi_1 F_N(\phi_2), \qquad \partial_{\phi_2} G_N = a^2(1-\phi_1^2)F_N'(\phi_2),$$
$$\partial_{\phi_3} G_N = 2aa'(\phi_3)(1-\phi_1^2)F_N(\phi_2). \tag{79}$$

In the notation of [7, Theorem 1], their transition and emission radii are, up to constants depending on the alphabet size and on their fixed spectral-gap margin L_A,

$$r_Q(T,x) = \frac{x(\delta_\mathrm{A} \vee \epsilon_\mathrm{A}\zeta_\mathrm{A})}{\sqrt{T\delta_\mathrm{A}^2\epsilon_\mathrm{A}^4\zeta_\mathrm{A}^6}}, \qquad r_f(T,x) = \frac{x}{\sqrt{T\delta_\mathrm{A}^2\epsilon_\mathrm{A}^4\zeta_\mathrm{A}^4}}, \tag{80}$$

with tail probability at most e^{-x^2}. On a fixed interior box, $\delta_\mathrm{A}, \epsilon_\mathrm{A}, \zeta_\mathrm{A}, L_\mathrm{A}$ are fixed and $a'(\phi_3)$ is bounded. Moreover, for $|\phi_2| \le \rho < 1$, equations (74)–(75) give

$$\sup_{|\lambda|\le\rho}|F_N(\lambda)| \le C_\rho N, \qquad \sup_{|\lambda|\le\rho}|F_N'(\lambda)| \le \frac{2N}{(1-\rho)^2}.$$

The Jacobian of $(p, q, f_1, f_2) \mapsto (\phi_1, \phi_2, \phi_3)$ is bounded on this box, so

$$\|\widehat{\phi} - \phi\|_2 \le C_\square\{r_Q(T, x) + r_f(T, x)\} \le C_\square \frac{x}{\sqrt{T}}.$$

The mean-value theorem applied to (78)–(79) thus yields

$$|\widehat{v}_\infty - v_\infty| \le C_\square \frac{x}{\sqrt{T}}, \qquad |\widehat{G}_N - G_N| \le C_\square N \frac{x}{\sqrt{T}}. \tag{81}$$

Taking $x = \sqrt{\log(1/\alpha)}$ reproduces exactly the powers of T, N, and $\log(1/\alpha)$ in Corollary 15.

Thus the upper-rate statement on a fixed Markov interior box is a direct functional consequence of the AGN parameter bounds, after the displayed Bernoulli coarsening; there is no unbounded-gradient or nonlinear obstruction there. A raw-parameter lower bound cannot be propagated in reverse by the delta method: a least-favourable parameter direction may be tangent to a level set of the functional. Indeed, $\partial_{\phi_1} v_\infty = 0$ at $p = q$. Nevertheless, our amplitude pair has

$$\partial_a v_\infty = 2aA_\infty > 0, \qquad \partial_a G_N = 2aA_N > 0,$$

and applying the same two-point functional principle on that regular direction gives (60)–(61). AGN's published lower theorem is not a formal lower bound for the Gaussian subfamily, because its least-favourable multinomial pairs need not lie on the Gaussian curve; our Gaussian KL calculation supplies that missing restriction. This is a routine model-specific Le Cam application, not a new minimax rate. Accordingly, the fixed-interior $T^{-1/2}$ and $NT^{-1/2}$ claims are not presented here as original consequences. AGN state their minimax lower radius at fixed probability 1/4; the explicit $\sqrt{\log(1/\alpha)}$ lower-confidence calibration in Corollary 15 comes from the Bretagnolle–Huber calculation above, not from a reverse delta argument.

Remark 19 (The AGN comparison on the small-transition front). AGN's Section II identifies three routes to the i.i.d. boundary, the first being $p \approx 0$ or $q \approx 0$. Their Theorem 1 and Corollary 1 give non-asymptotic bounds with explicit dependence on the lower margin $\delta_{\mathrm{A}} \le \min(p, q)$, so the small-transition front is covered by their published result [7, Section II, Theorem 1 and Corollary 1].

The precise scope still requires their spectral-gap qualification. They work on

$$\Theta_{L_{\mathrm{A}}} = \Theta(\delta_{\mathrm{A}}, \epsilon_{\mathrm{A}}, \zeta_{\mathrm{A}}) \cap \{1 - |1 - p - q| \ge L_{\mathrm{A}}\},$$

treat L_{A} as fixed, and state in Remark 1 that this permits one of p, q to vanish, but not both. For the joint symmetric path $p = q = u \downarrow 0$, membership requires $L_{\mathrm{A}} \le 2u$; hence no single fixed-L_{A} class contains the whole path. The point (u, u) is covered for every fixed $u > 0$ after choosing $L_{\mathrm{A}} \le 2u$. Their theorem statement suppresses the dependence of $C(K, L_{\mathrm{A}})$, but their proof of Lemma 1 permits the explicit tracking carried out below. AGN themselves state that accurate rates without a spectral-gap lower bound are beyond their scope and that tracking the gap with their method leaves an upper/lower mismatch by a factor δ_{A}. The distinction is visible directly from

$$\mathrm{Cov}\{\mathbf{1}_{\{S_0=2\}}, \mathbf{1}_{\{S_h=2\}}\} = \frac{pq}{(p+q)^2}(1 - p - q)^h.$$

If $p \downarrow 0$ and $q = q_0 > 0$, this covariance tends to zero; if $p = q = u \downarrow 0$, it tends to 1/4 for every fixed h. Thus the one-sided front approaches a one-component i.i.d. law, whereas simultaneous slow mixing does not.

We now quantify that dependence. Following [7, proof of Lemma 1], set

$$m_L = \left\lceil \frac{\log 4}{L} \right\rceil + 2, \qquad \gamma_L = \frac{1}{2m_L}. \tag{82}$$

Their length-three window chain has mixing time at most m_L, and the pseudo-spectral-gap comparison used in their proof gives $\gamma_{\text{ps}} \geq \gamma_L$. Solving their displayed Bernstein inequality coordinatewise and taking a union bound over the K^3 cells gives the admissible explicit radius

$$r_3(T, y; L) \leq C_K \left\{ \sqrt{\frac{y}{T\gamma_L}} + \frac{y}{T\gamma_L} \right\}, \qquad y = \log \frac{2K^3}{\alpha}. \tag{83}$$

This is the dependence hidden in their fixed-L notation; it is an explicit admissible tracking of their proof, not a claim that their constants are optimal.

Fix $\epsilon_0, \zeta_0 > 0$, put

$$\delta_{\text{A}} = u, \qquad \epsilon_{\text{A}} = \epsilon_0 \leq |1 - 2u|, \qquad \zeta_{\text{A}} = \zeta_0, \qquad L_{\text{A}} = 2u,$$

and assume $u < \epsilon_0\zeta_0$. The inverse factor in AGN's Theorem 1 is

$$\frac{\delta_{\text{A}} \vee \epsilon_{\text{A}}\zeta_{\text{A}}}{\delta_{\text{A}}\epsilon_{\text{A}}^2\zeta_{\text{A}}^3} = \frac{1}{u\epsilon_0\zeta_0^2}.$$

Moreover,

$$\gamma_{2u} = \frac{1}{2\{\lceil(\log 4)/(2u)\rceil + 2\}} \asymp u. \tag{84}$$

Substitution of (83) therefore yields the fully tracked AGN sufficient radius

$$r_Q^{\text{AGN}}(u, T, \alpha) \leq C_{K,\epsilon_0,\zeta_0} \left\{ \frac{\sqrt{y}}{u^{3/2}\sqrt{T}} + \frac{y}{u^2 T} \right\}. \tag{85}$$

Thus tracking $L_{\text{A}} = 2u$ adds the factor $u^{-1/2}$ which was hidden when $C(K, L_{\text{A}})$ was held fixed.

This is not yet a rate for v_∞. At $p = q = u$,

$$v_\infty(u, a) = \sigma^2 + a^2\frac{1-u}{u}, \qquad \partial_p v_\infty = \partial_q v_\infty = -\frac{a^2}{2u^2}, \qquad \frac{d}{du}v_\infty(u, a) = -\frac{a^2}{u^2}. \tag{86}$$

Fixing a and applying the ordinary delta upper bound in the local regime $r_Q^{\text{AGN}} = o(u)$ therefore gives

$$\begin{aligned} r_{v,\text{delta}}^{\text{AGN}} &\leq (|\partial_p v_\infty| + |\partial_q v_\infty|)\, r_Q^{\text{AGN}} + O\{(r_Q^{\text{AGN}})^2/u^3\} \\ &= O\left\{a^2\left(\frac{\sqrt{y}}{u^{7/2}\sqrt{T}} + \frac{y}{u^4 T}\right)\right\}. \end{aligned} \tag{87}$$

For $Nu \gg 1$, the same calculation gives $r_{G_N,\text{delta}}^{\text{AGN}} = O(N r_{v,\text{delta}}^{\text{AGN}})$. The delta propagation adds another u^{-2}. The leading tracked AGN functional exponent is therefore $u^{-7/2}$, whereas the local long-lag upper bound in Theorem 12 has exponent $u^{-3/2}$.

Lemma 20 (Exact KL control on the symmetric boundary path)**.** *Let $\overline{P}_u^{(T)}$ and $P_u^{(T)}$ denote respectively the complete-data and observed-data laws in the symmetric Markov model*

$$Q_u = \begin{pmatrix} 1-u & u \\ u & 1-u \end{pmatrix},$$

with stationary initialization and the same fixed Gaussian emissions under both alternatives. If $T \geq 2$, $0 < u \leq 1/4$, and $0 < d \leq u/2$, then

$$\begin{aligned} D_{\text{KL}}\left(P_u^{(T)} \middle\| P_{u+d}^{(T)}\right) &\leq D_{\text{KL}}\left(\overline{P}_u^{(T)} \middle\| \overline{P}_{u+d}^{(T)}\right) \\ &= (T-1)\,\text{kl}(u \| u+d) \\ &\leq \frac{(T-1)d^2}{(u+d)(1-u-d)} \leq \frac{8(T-1)d^2}{5u}. \end{aligned} \tag{88}$$

Proof. Put $v = u+d$. The stationary distribution is $\pi_u = \pi_v = (1/2, 1/2)$, and the conditional Gaussian laws are unchanged. The chain rule therefore gives the exact calculation

$$\begin{aligned} D_{\mathrm{KL}}\left(\overline{P}_u^{(T)} \middle\| \overline{P}_v^{(T)}\right) &= D_{\mathrm{KL}}(\pi_u \| \pi_v) + \sum_{t=1}^{T-1} \mathbb{E}_u\left[\log \frac{Q_u(S_t, S_{t+1})}{Q_v(S_t, S_{t+1})}\right] \\ &\quad + \sum_{t=1}^{T} \mathbb{E}_u\left[\log \frac{g_{S_t}(X_t)}{g_{S_t}(X_t)}\right] \\ &= (T-1)\left\{u \log \frac{u}{v} + (1-u)\log\frac{1-u}{1-v}\right\} \\ &= (T-1)\,\mathrm{kl}(u\|v). \end{aligned}$$

Indeed, under either transition row the switch indicator is $\mathrm{Bern}(u)$. Projection from the complete path onto the observations and the data-processing inequality give the first inequality in (88). Finally,

$$\mathrm{kl}(u\|u+d) \le \chi^2\{\mathrm{Bern}(u)\|\,\mathrm{Bern}(u+d)\} = \frac{d^2}{(u+d)(1-u-d)}.$$

Since $u + d \ge u$ and $1 - u - d \ge 5/8$, the final inequality follows. $\square$

The Fisher expansion is an interpretation of (88), not its justification. If $f_u(d) = \mathrm{kl}(u\|u+d)$, then

$$f_u(0) = f_u'(0) = 0, \qquad f_u''(0) = \frac{1}{u(1-u)},$$

and Taylor's theorem gives, for $0 \le d \le u/2$,

$$\mathrm{kl}(u\|u+d) = \frac{d^2}{2u(1-u)} + O\left(\frac{d^3}{u^2}\right). \tag{89}$$

More explicitly,

$$f_u'''(d) = -\frac{2u}{(u+d)^3} + \frac{2(1-u)}{(1-u-d)^3}, \qquad \sup_{0\le d\le u/2} |f_u'''(d)| \le \frac{C}{u^2},$$

so the ratio of the Taylor remainder to the quadratic term is $O(d/u)$. The quadratic approximation is therefore relatively accurate only when $d/u \to 0$. At the Fisher scale $d \asymp \sqrt{u(1-u)/T}$, this condition is $Tu \to \infty$. The earlier working draft stated the resulting $u^{-3/2}T^{-1/2}$ expression without this regime condition; that was an invalid extrapolation of the Taylor approximation and is corrected below.

The lower bound itself can be made uniform without Taylor expansion. Fix $0 < c_0 \le 1/2$ and set

$$d_{T,u} = c_0 \min\left\{u, \sqrt{\frac{u(1-u)}{T-1}}\right\}. \tag{90}$$

Lemma 20 gives

$$D_{\mathrm{KL}}\left(P_u^{(T)} \middle\| P_{u+d_{T,u}}^{(T)}\right) \le \frac{8}{5}c_0^2 \min\{(T-1)u, 1-u\} \le \frac{8}{5}c_0^2.$$

Moreover, the functional separation is exact:

$$|v_\infty(u + d_{T,u}, a) - v_\infty(u, a)| = \frac{a^2 d_{T,u}}{u(u + d_{T,u})}.$$

The two-point inequality therefore implies, for every estimator,

$$\max_{\vartheta\in\{u,u+d_{T,u}\}} P_\vartheta^{(T)}\left\{|\widehat{v}_{\infty,T} - v_\infty(\vartheta,a)| \geq \frac{a^2 d_{T,u}}{2u(u+d_{T,u})}\right\}$$
$$\geq \frac{1}{4}\exp\left(-\frac{8}{5}c_0^2\right). \tag{91}$$

Since $u + d_{T,u} \leq 3u/2$, this construction yields the two-point lower-bound scale

$$a^2 \min\left\{\frac{1}{u}, \frac{\sqrt{1-u}}{u^{3/2}\sqrt{T-1}}\right\}. \tag{92}$$

Thus $a^2u^{-3/2}T^{-1/2}$ is the $Tu \gg 1$ branch. When $Tu = O(1)$, the legitimate boundary branch is a^2/u, not an extrapolation of the Fisher formula.

Corollary 21 (Confidence-calibrated persistence lower radius). *Let* $0 < \delta < 1/4$, $\ell_\delta = \log\{1/(4\delta)\}$, *and*

$$d_{T,u,\delta} = \min\left\{\frac{u}{2}, \sqrt{\frac{5u\ell_\delta}{8(T-1)}}\right\}. \tag{93}$$

Then every estimator satisfies

$$\max_{\vartheta\in\{u,u+d_{T,u,\delta}\}} P_\vartheta^{(T)}\left\{|\widehat{v}_{\infty,T} - v_\infty(\vartheta,a)| \geq \frac{a^2 d_{T,u,\delta}}{2u(u+d_{T,u,\delta})}\right\} \geq \delta, \tag{94}$$

and the threshold on the left is at least

$$\frac{a^2}{6}\min\left\{\frac{1}{u}, \frac{\sqrt{\ell_\delta}}{u^{3/2}\sqrt{T-1}}\right\}. \tag{95}$$

Proof. Equation (88) and (93) give

$$D_{\mathrm{KL}}\left(P_u^{(T)}\middle\|P_{u+d_{T,u,\delta}}^{(T)}\right) \leq \ell_\delta.$$

Bretagnolle–Huber therefore gives $(1/4)e^{-\ell_\delta} = \delta$. Since $u + d_{T,u,\delta} \leq 3u/2$, the functional threshold is at least $a^2 d_{T,u,\delta}/(3u^2)$, which yields (95). ☐

For $u \leq 1/6$, take $u_\star = u$, $\alpha = 1$, and $\beta = 3/2$ in Theorem 12; the lower pair in Corollary 21 lies in that same local cell. For fixed known a, σ with non-degenerate signal-to-noise ratio, the upper and lower radii have identical powers:

	lower radius	upper radius
$Tu \gg \log(1/\delta)$	$a^2u^{-3/2}\sqrt{\log(1/\delta)/T}$	$(\lvert a\rvert+\sigma)^2u^{-3/2}\sqrt{\log(1/\delta)/T}$
$Tu = O\{\log(1/\delta)\}$	a^2u^{-1}	a^2u^{-1}

Thus the persistent-path result is minimax-rate tight on a local cell known up to a fixed multiplicative factor. The regular branch is a genuine statistical matching result. The boundary branch is diameter saturation and does not imply consistent relative estimation. This is a local minimax statement indexed by the prescribed scale $u_\star$. Since the estimator itself depends on $u_\star$, it does not imply a single estimator attaining these radii uniformly over unknown persistence scales. No global adaptive minimax claim is made.

This calculation does not assume local asymptotic normality of the observed likelihood. It uses an exact complete-data identity followed by data processing, so any singularity of the observed HMM can only make the observed KL smaller. The price is possible looseness: a complete-data KL upper bound need not reveal the sharp observed-data geometry. AGN's reparametrization and Proposition 2 are needed precisely to control that sharper marginal geometry near an i.i.d. boundary. In their notation $\phi_2 = 1 - p - q$; Proposition 2 assumes $|\phi_2|$ small, whereas on the symmetric slow-mixing path $\phi_2 = 1 - 2u \to 1$. The two KL calculations therefore concern different degeneracies.

Finally, $d_{T,u} \downarrow 0$ is a separation used to prove an impossibility result, not by itself an achievable estimation radius. In the $Tu \gg 1$ branch its relative size is

$$\frac{d_{T,u}}{u} \asymp \frac{1}{\sqrt{Tu}},$$

which worsens as u decreases. Omitting confidence and fixed signal-to-noise factors, the explicit comparison is

	radius for u	radius for v_∞
AGN tracked uniform upper	$u^{-3/2}T^{-1/2}$	$a^2u^{-7/2}T^{-1/2}$
long-lag local upper, $Tu \gg 1$	$u^{1/2}T^{-1/2}$	$a^2u^{-3/2}T^{-1/2}$
two-point local lower, $Tu \gg 1$	$u^{1/2}T^{-1/2}$	$a^2u^{-3/2}T^{-1/2}$
local boundary, $Tu = O(1)$	u	a^2u^{-1}

The tracked AGN radius is looser by u^{-2} on the symmetric Gaussian submodel. Their proof first estimates a fixed length-three law at the small-gap scale $(Tu)^{-1/2}$ and then applies an inverse bound of order u^{-1}. In contrast, the long-lag moment has the same $(Tu)^{-1/2}$ fluctuation but its inverse derivative is of order u, giving $\sqrt{u/T}$. The discrepancy is therefore attributable to the fixed-block uniform AGN procedure and its broader parameter class, not to an inherent lower bound on the present local Gaussian subexperiment.

Nor can (85) be extrapolated at fixed T all the way to $u = 0$: AGN's stated confidence range requires $y \le Tu^2\epsilon_0^4\zeta_0^6$, and their theorem treats L_{A} as fixed, while the symmetric path has $L_{\mathrm{A}} = 2u$. Theorem 12 supplies the matching observed-data upper bound for (92) on a prescribed local multiplicative cell. This comparison supplies no global adaptive theorem because the lag scale is fixed in advance.

For comparison, the fixed-L_{A} small-transition front covered directly by AGN may be represented by $p = u \downarrow 0$ and $q = q_0 > 0$ fixed. Then L_{A} remains bounded away from zero and

$$v_\infty(p, q_0, a) - \sigma^2 = \frac{4a^2pq_0(2-p-q_0)}{(p+q_0)^3} = \frac{4a^2(2-q_0)}{q_0^2}p + O(p^2).$$

Here the functional derivative stays finite, so the AGN $\delta_{\mathrm{A}}^{-1}T^{-1/2}$ raw radius propagates with the same δ_{A}^{-1} power. This is distinct from sending both hazards to zero.

Finally, for finite D, Markovization produces $2D$ latent age states with identical emission laws across the D ages of each regime. AGN's two-state theorem does not apply to that augmented representation, and the repeated emission columns preclude treating it as an unrestricted identifiable $2D$-state HMM.

Discounted Markov-cost variance. To avoid the collision of notation with this article, write B for the total transition budget, H for the truncation horizon, and $\rho \in [0, 1)$ for the discount factor in [10]. Their target and truncated return are

$$D_\rho = \sum_{t=0}^{\infty} \rho^t f(S_t), \qquad D_{\rho,H} = \sum_{t=0}^{H-1} \rho^t f(S_t).$$

Their reset access produces $m = \lceil B/H \rceil$ independent copies of $D_{\rho,H}$; states and costs are observed, and $|f(s)| \le K$. In the published version, Theorem 4.12 states

$$\mathbb{E}|\widehat{V}_B - \mathrm{Var}(D_\rho)| \le \frac{8(1-\rho^H)^2 K^2}{\sqrt{m}(1-\rho)^2} + \frac{4\rho^H K^2}{(1-\rho)^2}, \tag{96}$$

$$\mathbb{P}\{|\widehat{V}_B - \mathrm{Var}(D_\rho)| > \varepsilon\} \le 2\exp\left[-\frac{m(1-\rho)^4}{32(1-\rho^H)^4 K^4}\left\{\varepsilon - \frac{4\rho^H K^2}{(1-\rho)^2}\right\}^2\right] \tag{97}$$

for $\varepsilon > 4\rho^H K^2/(1-\rho)^2$. Equivalently, its $(1-\alpha)$-confidence radius is

$$r_{\mathrm{TPB}}(B, H, \alpha) = \frac{4\rho^H K^2}{(1-\rho)^2} + \frac{\sqrt{32}(1-\rho^H)^2 K^2}{(1-\rho)^2}\sqrt{\frac{\log(2/\alpha)}{m}}. \tag{98}$$

Thus Theorem 4.12 is an upper concentration result for an episodic sample variance, not a lower bound for a hidden single-trajectory experiment.

The comparable horizon limit in our notation is $N \to \infty$, not $T \to \infty$: our T is data length, whereas their H, denoted T in their paper, is an estimator truncation parameter. If $c_h = \mathrm{Cov}(X_0, X_h)$ and $\sum_{h \ge 0} |c_h| < \infty$, then

$$V_\rho := \mathrm{Var}\left(\sum_{t=0}^{\infty} \rho^t X_t\right) = \frac{c_0 + 2\sum_{h=1}^{\infty} \rho^h c_h}{1-\rho^2}, \tag{99}$$

$$(1-\rho^2)V_\rho \longrightarrow v_\infty, \qquad \frac{G_N}{N} \longrightarrow v_\infty. \tag{100}$$

Hence, with the effective horizon $N_\rho = (1-\rho^2)^{-1}$, one has the population equivalence $V_\rho \sim G_{N_\rho}$. This is the only direct convergence between the two formulations.

Indeed, set $h = 1-\rho$ and normalize the discounted target as $W_\rho = (1-\rho^2)V_\rho$. Multiplying (98) by $1-\rho^2$ gives

$$\begin{aligned} r_W(B, H, \alpha) = &\frac{4\rho^H K^2(1+\rho)}{h} \\ &+ \frac{\sqrt{32}(1-\rho^H)^2 K^2 (1+\rho)}{h}\sqrt{\frac{\log(2/\alpha)}{m}}. \end{aligned} \tag{101}$$

To make the first term at most $\eta/2$, it is necessary that

$$\rho^H \le \frac{\eta h}{8K^2(1+\rho)}, \qquad H \ge \frac{\log\{8K^2(1+\rho)/(\eta h)\}}{-\log\rho} \sim \frac{\log(1/h)}{h}. \tag{102}$$

Once this truncation bias is controlled, the stochastic term in (101) requires

$$m \gtrsim \frac{K^4}{\eta^2 h^2}\log\frac{2}{\alpha}.$$

Since $B \asymp mH$, the resulting sufficient budget obeys

$$B \gtrsim \frac{K^4}{\eta^2 h^3}\log\frac{2}{\alpha}\log\frac{K^2}{\eta h}. \tag{103}$$

Thus the normalized Theorem 4.12 bound is singular as $\rho \uparrow 1$ and does not converge to the $\sqrt{\log(1/\alpha)/T}$ bound for v_∞. The discrepancy survives the population normalization because TPB uses independent reset episodes, a generic bounded-cost sample-variance estimator, and an explicit truncation bias, whereas the present experiment consists of one stationary hidden trajectory with unbounded Gaussian emissions and an exact parametric functional. Their general lower bound already establishes the root-sample barrier for variance risk; consequently that power is not new here either. The present result is therefore not merely a finite-horizon extension of Theorem 4.12. Its distinct scope is the explicit functional separation for G_N and v_∞ in the stationary hidden capped-duration experiment.

8.6 Scope of the optimality claim

On the fixed interior classes, the upper and lower bounds match at the minimax-rate level; the global Doeblin and inversion constants are not claimed sharp. Along the persistence path, Theorem 12 and Corollary 21 match powers cell by cell, but the lag uses the prescribed scale $u_\star$. We do not prove adaptation over a union of unknown scales. For unbounded durations controlled only through two moments, we make no uniform exponential claim; such a result would require stronger tail assumptions than are imposed here.

9 Numerical audit of the persistent-path rates

This section checks the finite-sample indices and constants of Theorem 12 and Corollary 21; it is not used in either proof. For $W_r = X_{1+rk}X_{1+(r+1)k}/a^2$, put $\rho = (1-2u)^k$ and $s = \sigma/|a|$. Direct calculation gives

$$\mathrm{Var}(W_r) = (1+s^2)^2 - \rho^2, \qquad \mathrm{Cov}(W_r, W_{r+1}) = s^2\rho^2, \qquad \mathrm{Cov}(W_r, W_{r+j}) = 0 \quad (j \geq 2). \tag{104}$$

Consequently the exact finite-M benchmark is

$$\mathrm{Var}(\widehat{R}_k) = \frac{M\{(1+s^2)^2 - \rho^2\} + 2(M-1)s^2\rho^2}{M^2}. \tag{105}$$

The derivative used for the finite-sample delta benchmark is

$$\left|\frac{dv_\infty}{dR_k}\right| = \frac{a^2(1-2u)^{1-k}}{2ku^2}. \tag{106}$$

We used $a = \sigma = 1$, $T = 65537$, 2500 independent stationary trajectories at each

$$u \in \{0.0400, 0.0300, 0.0225, 0.0169, 0.0127, 0.0095\},$$

and the prescribed cell $\mathcal{U}_\star = [u/4, 4u]$, with $u_\star = u$. The implementation simulated the lag-one and long-lag statistics from the same trajectory. Table 1 reports regressions of the logarithm of RMSE on $\log u$. The moment standard-deviation ratio is the Monte Carlo standard deviation divided by (105); the last column compares the nonlinear plug-in RMSE with the delta standard deviation.

Table 1: Finite-sample power and variance audit.

Statistic	Empirical slope	Delta slope	Moment SD ratio	RMSE/delta SD
Long-lag plug-in	−1.551	−1.497	0.999–1.019	1.051–1.139
Lag-one linearization	−2.007	−2.008	0.983–1.012	—
Lag-one nonlinear plug-in	−2.495	−2.008	0.983–1.012	1.041–2.007

Thus the long-lag exponent agrees with $-3/2$, while the linearized lag-one exponent agrees with -2. The nonlinear lag-one plug-in leaves its delta regime at the smallest hazards, which steepens its finite-grid slope; this effect is visible rather than removed by post-selection. The long-lag RMSE crosses below the lag-one RMSE between $u = 0.0300$ and $u = 0.0225$; at $u = 0.0095$ the respective RMSE values are 35.97 and 96.78.

For the saturation audit, take $u_0 = 0.01$, $\delta = 0.05$, $\ell = \log\{1/(4\delta)\} = \log 5$, the cell $[u_0, 3u_0/2]$, and 10000 replications under each member of the pair $(u_0, u_0 + d)$, where

$$d = \min\left\{\frac{u_0}{2}, \sqrt{\frac{5u_0\ell}{8(T-1)}}\right\}.$$

The two branches meet exactly at

$$(T-1)u_0 = \frac{5}{2}\ell = 4.024\ldots. \tag{107}$$

Across $1 \le (T-1)u_0 \le 5120$, the largest exact complete-data KL-to-ℓ ratio was 0.3128, so the smallest exact Bretagnolle–Huber probability was 0.1511, above the required 0.05. At the half-separation threshold, the smallest observed value of the maximum failure probability over the two alternatives was 0.4777.

The algebraic switch in (107) should not be confused with the point at which the projected estimator visibly leaves its diameter plateau. When $a = \sigma = 1$, $k \sim u^{-1}$, and $Tu \to \infty$, (105)–(106) give

$$\frac{u}{a^2}\,\mathrm{sd}(\widehat{v}_{\infty,k}) \sim \frac{C_{\mathrm{L}}}{\sqrt{Tu}}, \qquad C_{\mathrm{L}} = \frac{e^2}{2}\sqrt{4+e^{-4}} = 7.406\ldots. \tag{108}$$

At 90% probability, equating $z_{0.95}C_{\mathrm{L}}/\sqrt{Tu}$ to the normalized cell diameter $1/3$ predicts

$$Tu \simeq \{3z_{0.95}C_{\mathrm{L}}\}^2 = 1335.5.$$

The simulated 90% error quantile equalled the diameter through $Tu = 640$ and first fell below it at $Tu = 1280$. The proximity of the simulated and predicted thresholds checks the normalization and the indexing of the saturation calculation, but no more: the theorem proves the rate match, not optimality of its leading constant.

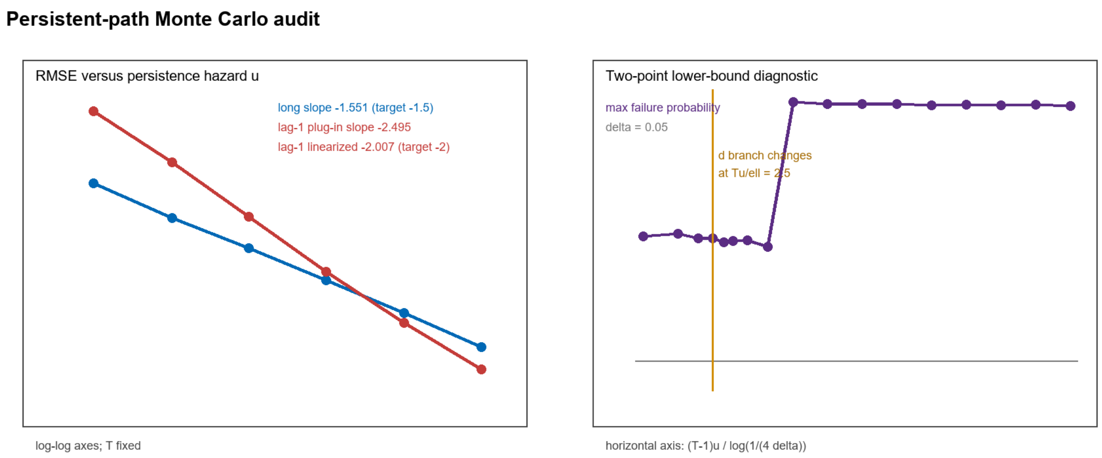

Figure 1: Left: fixed-T power audit for long lag and lag one. Right: the empirical two-point failure probability and the exact branch-switch location.

The simulation is reproducible from `Paper/simulacion_reproducible_v2.py`; the complete numerical output is stored in the two versioned CSV files in `Paper`.

Even with observed states, the number of complete cycles is approximately T/T_c; duration parameters therefore have a baseline error scale $\sqrt{T_c/T}$, before dimension and hidden-state separation are included. The mean cycle length alone is not a mixing constant: residual-life tails, minimum duration probabilities, and the gap of Q_F also matter.

If F_i is arbitrary with only finite mean and variance, an exponential Bernstein bound cannot hold uniformly. Exponential or bounded duration tails are sufficient routes to geometric mixing. Under polynomial tails one must use regenerative/Fuk–Nagaev-type bounds, and estimating ν_i^2 at a root-number-of-cycles rate generally requires a fourth duration moment. Likewise, finite emission variance alone is insufficient for exponential concentration of empirical second moments.

10 Prior-art map and scope of the contribution

The following distinctions are important.

1. Nonparametric HMM identification under full-rank transitions and linearly independent emissions is established in [3]. Spectral/moment algorithms with finite-sample guarantees are established for identifiable HMM classes in [4, 5]; nonparametric oracle/minimax inequalities appear in [6, 7, 11].

2. Bernstein, empirical-process, and regenerative inequalities for additive functionals of geometrically ergodic Markov chains are already available in [13, 14, 15, 12]. The inequality actually used here is Paulin [12, Theorem 3.11, equation (3.25), corrected author version]: it treats a stationary, not necessarily reversible chain satisfying Assumption 3.1 on a Polish state space, a bounded $L^2(\pi)$ summand, and a positive pseudo-spectral gap. Its Proposition 3.4, equation (3.9), converts uniform-ergodicity mixing time into the pseudo-spectral-gap lower bound used in the non-asymptotic and explicit capped-duration sections. Since a bounded-duration HSMM becomes Markov after residual-time augmentation, and a lag product becomes additive after window augmentation, concentration of empirical autocovariances is covered at this generic level.

3. Merlevède, Peligrad and Rio [16, Theorem 2] give the corresponding strong-mixing antecedent for centered, uniformly bounded variables with $\alpha(k) \leq e^{-2ck}$; stationarity is not required. Their denominator contains the covariance proxy mv^2, a boundedness term, and a linear term multiplied by $(\log m)^2$. This result is not used to justify the log-free Paulin radius above.

4. Finite-sample error bounds for empirical autocovariance operators under strong mixing are available from Mollenhauer et al. [17]; their direct insertion into a HAC estimator is therefore background rather than a novelty claim.

5. For observed Markov-chain functionals, non-asymptotic long-run-variance estimation predates this work: Gyori and Paulin [18] analyze a truncated autocovariance estimator. We therefore make no general first-estimator claim.

6. Melnyk and Banerjee's spectral algorithm [19] handles discrete HSMMs with bounded duration and full-rank conditions, but from a set of independent training sequences and with the aim of scoring a test sequence—a different statistical experiment and target from the present one, which starts from one trajectory and targets G_N or v_∞ directly.

7. Barbu and Limnios [20] and Trevezas and Limnios [21] prove consistency and asymptotic normality of HSMM maximum-likelihood estimators. Farcomeni [22] derives exact score and information recursions. These are important computational and asymptotic antecedents, but not end-to-end high-probability functional bounds.

8. Khorshidian et al. [23] prove strong consistency and asymptotic normality for moments of semi-Markov reward processes, but from an observed semi-Markov kernel—the hidden-state identification problem from emissions, which is our starting point, does not arise in their setting.

9. Posterior concentration for semi-Markov kernels in Votsi et al. [24] concerns observed histories; hidden emissions and accumulated variance are outside that experiment.

10. De Castro, Gassiat and Le Corff [25] study filtering and smoothing in a nonparametric *HMM*; this paper is not an HSMM parameter-estimation theorem.

We do not rely on a joint result by Douc, Moulines, and Rio, and no such citation is retained. The concentration step for the Markov component is instead based on Paulin [12, Theorem 3.11], with its state-space, stationarity, boundedness, and pseudo-spectral-gap hypotheses stated in Section 5. Where the strong-mixing route is discussed, Merlevède, Peligrad and Rio [16, Theorem 2] are cited only for that statement. Neither reference is used to claim a concentration theorem for the present hidden semi-Markov functional; the functional bounds are proved under the assumptions stated in their respective theorems.

Theorem 10 gives the generic composition for the stated block-moment estimator. Section 6 makes this composition quantitative: for a named capped-geometric Gaussian family it gives an exact observable inverse, a closed uniform $\underline{\kappa}_{\mathrm{id,cg}}$, a closed uniform $\underline{g}_3$, and an end-to-end radius with Paulin's numerical constants. The restriction to antipodal means and known common variance is explicit; the result does not silently claim the four-free-emission-parameter model.

A matching lower bound and the sensitivity calculation for G_N and v_∞ are given in Section 8. On fixed interior HMM classes, the root-data-length parameter rates are already covered by Abraham, Gassiat and Naulet; propagation to the present smooth functionals is not claimed as a new generic rate. The upper and lower functional powers match in T, $\log(1/\delta)$, and N on the stated fixed interior box, but the uniform Doeblin and global inverse constants are not claimed to be sharp. On the symmetric persistent Markov path with fixed known emission parameters, the long-lag estimator of Section 7 matches the two-point lower bound in both the regular and saturated regimes on a local multiplicative cell. The precise restrictions on this comparison are collected in Section 8.6.

AI tool disclosure

OpenAI Codex (GPT-5 family, accessed through the Codex desktop application, most recently on 28 August 2026) assisted with mathematical exposition, bibliographic discovery, code generation, and manuscript editing. The author retains full responsibility for the mathematical statements, citations, computations, and accompanying code.

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